

# Project Risk Analysis

## Chapter Overview

In this chapter, we investigate the sources of *uncertainty* in forecasts of future cash flows. Although the presentation of project cash flows is based on accounting income statements, these are not historical statements. They are pro forma, or forecasted, statements and are thus highly uncertain. This chapter also describes the various approaches that are used to analyze risk and deal with uncertainty. Specifically, under the broad umbrella of sensitivity analysis, we consider scenario analysis, breakeven sensitivity analysis, and Monte Carlo simulation. Finally, we provide an initial discussion of the role that investment flexibility can have on expected cash flows and the role that decision trees can play in organizing the analysis of decision flexibility.

## 1 INTRODUCTION

The use of discounted cash flow (DCF) analysis to evaluate major long-term investments is now accepted practice throughout the world. In its simplest form, analyzing capital expenditures using DCF involves estimating a single stream of expected cash flows, discounting it, and then making a decision based on a single estimate of the investment's NPV. In practice, however, investment decisions take place in a world of uncertain future outcomes, where there are more things that *can* happen than *will* happen, which makes investment analysis considerably more complex.

In this chapter we explicitly confront the challenge of uncertainty and focus on investment risk analysis. In particular, we will consider various techniques that financial analysts use to learn more about an investment's primary value drivers and the risks they generate.

To understand how uncertainty is accounted for in an investment analysis, we find it helpful to think about how financial analysts evaluate new investment opportunities in terms of two phases:

- In *Phase I*, the analyst tries to envision the possible outcomes from an investment and to come up with an estimate of what he or she *thinks* or *expects* will happen. This analysis forms the basis for estimating the NPV, IRR, and other measures of investment value.

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- In *Phase II*, the analyst details the underlying sources of risk. This involves identifying the investment's value drivers and the uncertainty that characterizes each. Once these risks are identified, the analyst will seek ways to mitigate some of those risks while recognizing the firm's exposure to other risks and the need to monitor the risks carefully throughout the life of the project.

The challenge to the financial analyst in implementing DCF analysis, then, is twofold: First, the analyst must make all the forecasts and assumptions necessary to generate an estimate of the investment's NPV. Second, the analyst must perform an in-depth analysis of the assumptions used when estimating NPV to come to grips with what might happen to the project when (not if) things do not go as planned. In this chapter we describe a variety of tools that analysts can use to test and probe their estimates of the NPV of a proposed project. These tools include scenario analysis, breakeven sensitivity analysis, and Monte Carlo simulation.

The tools of investment analysis that we discuss in this chapter may give an impression of a high level of scientific precision. However, the underlying basis for using them is inherently subjective, as it relies on the judgment of the individual(s) performing the analysis. Realization of this fact is often disconcerting to “number-focused” financial analysts who look for answers in techniques and tools. So a key learning point of this chapter involves coming to grips with the inherently subjective nature of the investment evaluation process and the critical role that judgment and experience must play.<sup>1</sup>

## 2 UNCERTAINTY AND INVESTMENT ANALYSIS

Our discussion throughout this chapter focuses on **project valuation**—the evaluation of investments in projects that an individual or firm might consider. However, the tools and discussion apply equally well to **enterprise valuation**—the evaluation of acquisitions of entire firms.

### The Investment Process with Risky Cash Flows

The approach involves the use of a three-step valuation process:

1. Estimate the amount and timing of future cash flows for each year of the proposed investment's life.
2. Identify a risk-appropriate discount rate.
3. Calculate the present value of future cash flows. If the expected net present value (NPV) of the project's cash flows is positive after considering all the relevant cash inflows and outflows, then the firm should undertake the project.

While in theory this approach provides the best estimate of the value created by the investment project, in practice the cash flow estimates are at best educated guesses and at worst out-and-out fiction. Hence, the initial DCF analysis should be viewed as only the first phase of the valuation process. This calculation is followed by Phase II, in which the analyst performs a form of “exploratory surgery” on the initial estimate. The object of

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<sup>1</sup> We often advise our more “technique-oriented” students that if there were no need for human judgment, there would be no need for financial analysts!

this “follow-on” analysis is to explore the investment’s value drivers—those factors that are most critical to the project’s success. The analyst cannot eliminate uncertainty, but can better understand the relative sensitivity of the project’s NPV to different key variables.

As an illustration, consider the following investment opportunity in an organic fertilizer product being considered by CSM Inc. and described below.

### Example—The Earthilizer Proposal

CSM Inc. is considering an investment in a project that requires an initial investment of \$580,000 to produce and market a new organic fertilizer made from dairy-farm waste. The product, Earthilizer, has been developed by CSM’s agricultural division over the past three years in response to two factors: First, there is a growing demand from organic farmers and second, dairy farmers face increasingly more restrictive environmental rules regarding the disposal of cattle manure. In combination, CSM’s management feels that these forces of change present a very promising investment opportunity.

CSM has tested the product extensively as a liquid-fertilizer replacement for the more expensive and conventional chemical fertilizers that farmers and ranchers have used since the Second World War. The product will be marketed in a concentrated form, with 1 gallon of Earthilizer producing about 100 gallons of usable fertilizer.

Not only is the product wholly organic, but it is also cheaper to use than chemical fertilizers. Traditional chemical fertilizers are made from petroleum and cost approximately \$25 an acre for grass and hay production; Earthilizer utilizes dairy waste and costs less than \$20 per application, with no more applications required over the growing season.

CSM’s analysts estimate that an initial investment of \$580,000 will put the Earthilizer project in place. This investment is comprised of \$250,000 in working capital plus \$330,000 in plant and equipment.

### Estimating Earthilizer’s Project Free Cash Flows (FCFs)

If the project is approved, the agricultural division of CSM plans to begin construction on the new facilities immediately. The manufacture and distribution of Earthilizer will be “up and running” by the end of 2010, with the project’s first year of revenues in 2011. Table 1 contains a set of financial projections for the project. These consist of pro forma income statements (Panel a of Table 1), balance sheets (Panel b), and cash flows (Panel c) spanning the five years used by CSM to evaluate the investment. Of course, if the project is successful, it could operate for many years. However, it is standard practice among many firms to select an arbitrary investment life and analyze the investment as if it will be shut down at the end of that period.

CSM Inc. is a very conservatively financed firm that finances all its investments out of internally generated cash flows (i.e., the firm does not borrow to obtain funds for new investments). Consequently, the project is all-equity-financed, as evidenced by the fact that there is no interest expense in the pro forma income statements (Panel a of Table 1), nor is there any debt financing in the pro forma balance sheets for the project (Panel b).

#### Did you know?

##### **Barnyard animals: An important source of green energy\***

Manure and other waste products from cows, pigs, and other livestock have long been recognized as a source of groundwater pollution. However, new technology for converting this waste to an energy source indicates it is a largely untapped source of energy in the United States. For example, fed by the waste of 900 cows, a Wisconsin dairy farm now produces about 6.5 million kilowatt hours a year—enough to power 600 homes annually. Moreover, meatpacker Swift & Co. recently signed a letter of intent to build a biogas production facility. The biogas from all seven of Swift’s plants has the potential to generate the equivalent of 25,000 barrels of heating oil a day.

\*[http://news.com/Manufacturing+power+from+manure/2009-11395\\_3-6057795.html](http://news.com/Manufacturing+power+from+manure/2009-11395_3-6057795.html)

**Table 1 CSM Inc. Earthilizer Investment Project****Panel a. Pro Forma Income Statements (\$000)**

| Income Statements                      | 2011       | 2012       | 2013       | 2014       | 2015       |
|----------------------------------------|------------|------------|------------|------------|------------|
| Sales (10% growth per year)            | \$1,000.00 | \$1,100.00 | \$1,210.00 | \$1,331.00 | \$1,464.10 |
| Cost of goods sold                     | (674.00)   | (741.40)   | (815.54)   | (897.09)   | (986.80)   |
| Gross profit                           | 326.00     | 358.60     | 394.46     | 433.91     | 477.30     |
| Operating expenses before depreciation | (215.00)   | (225.00)   | (236.00)   | (248.10)   | (261.41)   |
| Depreciation expense                   | (33.00)    | (33.00)    | (33.00)    | (33.00)    | (33.00)    |
| Earnings before interest and taxes     | 78.00      | 100.60     | 125.46     | 152.81     | 182.89     |
| Interest expense                       | —          | —          | —          | —          | —          |
| Earnings before taxes                  | 78.00      | 100.60     | 125.46     | 152.81     | 182.89     |
| Taxes                                  | (23.40)    | (30.18)    | (37.64)    | (45.84)    | (54.87)    |
| Net income                             | \$ 54.60   | \$ 70.42   | \$ 87.82   | \$ 106.96  | \$ 128.02  |

**Panel b. Pro Forma Balance Sheets (\$000)**

| Balance Sheets                 | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     |
|--------------------------------|----------|----------|----------|----------|----------|----------|
| Net working capital            | \$250.00 | \$250.00 | \$275.00 | \$302.50 | \$332.75 | \$366.03 |
| Gross plant and equipment      | 330.00   | 330.00   | 330.00   | 330.00   | 330.00   | 330.00   |
| Less: Accumulated depreciation | —        | (33.00)  | (66.00)  | (99.00)  | (132.00) | (165.00) |
| Net plant and equipment        | \$330.00 | \$297.00 | \$264.00 | \$231.00 | \$198.00 | \$165.00 |
| Total                          | \$580.00 | \$547.00 | \$539.00 | \$533.50 | \$530.75 | \$531.03 |

**Panel c. Expected Project Free Cash Flows (\$000)**

|                                          | 2010        | 2011     | 2012     | 2013     | 2014     | 2015     |
|------------------------------------------|-------------|----------|----------|----------|----------|----------|
| EBIT                                     | \$ —        | \$ 78.00 | \$100.60 | \$125.46 | \$152.81 | \$182.89 |
| Less: Taxes                              | —           | (23.40)  | (30.18)  | (37.64)  | (45.84)  | (54.87)  |
| NOPAT                                    | \$ —        | \$ 54.60 | \$ 70.42 | \$ 87.82 | \$106.96 | \$128.02 |
| Plus: Depreciation expense               | —           | 33.00    | 33.00    | 33.00    | 33.00    | 33.00    |
| Less: CAPEX                              | (330.00)    | —        | —        | —        | —        | —        |
| Less: Changes in net working capital     | (250.00)    | —        | (25.00)  | (27.50)  | (30.25)  | (33.28)  |
| Plus: Liquidation of net working capital |             |          |          |          |          | 366.03   |
| Plus: Liquidation of PPE                 |             |          |          |          |          | 165.00   |
| Equals: Project free cash flow (FCF)     | \$ (580.00) | \$ 87.60 | \$ 78.42 | \$ 93.32 | \$109.71 | \$658.77 |

Panel c of Table 1 details management's estimates of the project's free cash flows. Note that CSM's management expects the project to be cash flow positive in 2011, which is the first year of operations, and every year thereafter.

The cash flow calculations for 2011 to 2015 are based on a number of key estimates and assumptions:

- These estimates assume that the project will require an initial investment of \$580,000 at the end of 2010, which includes \$250,000 in net working capital and \$330,000 in property, plant, and equipment.
- Sales revenues are expected to be \$1,000,000 in 2011 and are expected to grow at an annual rate of 10% throughout the five-year planning period (Panel a).
- Cost of goods sold is equal to 67.40% of sales,<sup>2</sup> while operating expenses before depreciation are 10% of sales revenues plus a fixed component equal to \$115,000 per year (Panel a).
- The firm uses straight-line depreciation, where plant and equipment is assumed to have a 10-year life and a zero salvage value (Panels a and b).
- A 30% tax rate is used in the analysis (Panel a).
- For 2011 through 2012, the firm estimates that its net working capital requirements will equal 25% of Earthlizer's sales. CAPEX is assumed to be \$330,000 in 2010 and zero in all future years<sup>3</sup> (Panel c).

CSM assumes that it will terminate the project in 2015, at which time it will produce a terminal cash flow of \$658,770. This terminal cash flow is equal to the sum of NOPAT plus depreciation expense for 2015 (i.e., \$128,020 + 33,000 = \$161,020) plus the liquidating value (which is expected to equal the book value) of the firm's investment in net working capital (i.e., \$366,030 in Panel b) and the book value of plant and equipment at the end of 2015 (i.e., \$165,000 in Panel b).<sup>4</sup>

Of course, the project may not operate for the full five years, or it may operate for more years. Our analysis follows industry practice, which "assumes" a termination date for the project and estimates the cash flows that would be realized from either liquidating or selling the business. However, to simplify our analysis of the Earthlizer investment, we assume that the terminal value for the Earthlizer assets is equal to their liquidation value (i.e., working capital plus net property, plant, and equipment—PPE), which we assume to be equal to their book value.

### Valuing Earthlizer's Project FCFs

Our estimates of the expected project free cash flows generated by the Earthlizer proposal are found in Panel c of Table 1. Now it is time to evaluate their present value. To do this, we discount the expected values of the FCFs using a discount rate that reflects

<sup>2</sup> Note that this ratio implies a gross profit margin of  $32.6\% = 100\% - 67.4\%$ .

<sup>3</sup> Note that we are assuming that no new plant capacity is required over the five-year planning period. CSM simply spends \$330,000 in 2010.

<sup>4</sup> We assume that when the project is shut down, CSM will realize a cash flow equal to the book values of its investments in the project's net working capital and property, plant, and equipment. If the firm has better estimates of the values of these assets, then they (and the taxes on any gains or losses from those assets) should be used to estimate the terminal cash flow from the investment.

the operating risk characteristics of the project and the fact that the project is all-equity-financed (unlevered). For now, we assume that the appropriate (risk-adjusted) discount rate for the unlevered FCFs is 13.25%.<sup>5</sup> Using this rate to discount the FCFs found in Panel c of Table 1, we estimate the present value of the *expected project free cash flows* to be \$623,070:

$$\begin{aligned} \$623,070 = & \frac{\$87,600}{(1 + .1325)^1} + \frac{\$78,420}{(1 + .1325)^2} + \frac{\$93,320}{(1 + .1325)^3} \\ & + \frac{\$109,710}{(1 + .1325)^4} + \frac{\$658,770}{(1 + .1325)^5} \end{aligned}$$

### The Decision to Invest (or Not)—NPV and IRR

If the proposed project is worth \$623,070, should it be undertaken? The quick answer is yes. We have estimated a value for the project's future cash flows that exceeds the \$580,000 initial cost associated with making the investment. In net present value (NPV) terms, the project offers an NPV equal to the difference in the expected value we have placed on it (its DCF value estimate equals \$623,070) and the cost of making the investment (\$580,000), or \$43,070.

A second metric that managers use to evaluate an investment is the project's *rate of return*. We define the *internal rate of return*, or *IRR*, as the discount rate that will result in a zero NPV for the investment. In the above example, we estimated the IRR to be 15.36%. Hence, if the required rate of return is 13.25%, the investment earns an excess rate of return of 2.11%, which, like the positive NPV calculated above, suggests that the investment should be accepted.<sup>5</sup>

It should be emphasized, however, that this analysis does not mean that CSM is *guaranteed* a net present value of \$43,070 or that the return on the investment will be 15.36%. Because this is a risky project and things will not go exactly as planned, CSM needs to learn more about the project to get a better understanding of how confident they should be about the NPV estimate.

## 3 SENSITIVITY ANALYSIS—LEARNING MORE ABOUT THE PROJECT

Phase I of our analysis is now complete: We have an estimate of the NPV of the Earthilizer investment. But how confident can we be that the project will unfold as we expect? What are the key value drivers of the project that the firm should monitor over the life of the investment to ensure its success? We now enter Phase II, where we use a variety of tools to address these concerns. Specifically, we will discuss three such tools: scenario analysis, breakeven sensitivity analysis, and simulation analysis.

<sup>5</sup> We noted that the IRR is not always unique, because there can be as many IRRs as there are changes in signs in the cash flows. For the Earthilizer project, there is only one change in sign (i.e., the initial outlay is negative, and all the remaining Project FCFs are positive); thus, there is only one possible IRR.



## Scenario Analysis

**Scenario analysis** is a technique that helps analysts explore the sensitivity of an investment's value under different situations or scenarios that might arise in the future. Here we use the term *scenario* to refer to different sets of assumptions about the realized values of each of the value drivers.

For example, we might ask what would happen to the value of the Earthlizer project if the initial sales level were to drop to the marketing department's most pessimistic estimate of only \$500,000. In this case, the DCF value of the project would drop to \$342,790 and the project would become a negative-NPV investment, because the investment requires an initial commitment of \$580,000 (i.e., the project NPV would drop to  $\$342,790 - \$580,000 = -\$237,210$ ). We could also analyze scenarios involving *multiple sets* of changes in assumptions and forecasts. For example, we might evaluate the project first using the most optimistic estimates for the value drivers and then the most pessimistic estimates. Although scenario analysis is very useful, there is no systematic way to define the scenarios. The number of possible scenarios is limited only by the imagination of the analyst performing the analysis. One approach that is often used to systematize the sensitivity analysis is something called *breakeven sensitivity analysis*.

## Breakeven Sensitivity Analysis

**In breakeven sensitivity analysis**, we ask the question: "What is the critical value of the particular value driver that pushes the NPV down to zero?" Although we could use the trial-and-error analysis of a spreadsheet model to determine the critical values that answer this question, it is much easier to use the Goal Seek function in Excel.<sup>6</sup>

### BEHAVIORAL INSIGHT

#### Scenario Analysis As a Strategic Planning Tool at Shell

Shell Oil (RDS) found that scenario analysis was a valuable exercise to help its executives avoid thinking that was constrained by what Warren Buffet refers to as "the institutional imperative." This is the tendency for firms to resist change even in the face of overwhelming evidence that pursuing the current strategy is misguided.

Scenario analysis can enable key executives to prepare a carefully crafted story about the future. The process of writing such a story provides an opportunity for executives to learn about what can happen to an investment and, in so doing, to prepare

for the future accordingly. The idea is not so much that one becomes a better forecaster. Instead, the value of the process is that managers will make themselves aware of potential problems and opportunities so that they can be prepared to take advantage of them when and if they arise. The very act of identifying conditions under which the plug should be pulled on a project can make managers aware of when and why it should be done.

*Sources:* Shell.com, March 2004, and P. de Geus, "Planning Is Learning," *Harvard Business Review* (March–April 1988).

<sup>6</sup> Excel contains a tool called "Goal Seek" under the "Tools" menu that makes this type of analysis very easy to do once the cash flows and NPV have been modeled. See the Technical Insight box two pages ahead.

**Table 2** Breakeven Analysis of the Assumptions and Forecasts Underlying the CSM Inc. Earthlizer Project

| Variable—Forecast or Assumption<br>(1)          | Expected<br>Value<br>(2) | Critical<br>Value*<br>(3) | % Change<br>(4) |
|-------------------------------------------------|--------------------------|---------------------------|-----------------|
| Sales growth rate                               | 10.00%                   | 5.66%                     | −43.40%         |
| <b>Gross profit margin = Gross profit/Sales</b> | <b>32.60%</b>            | <b>31.12%</b>             | <b>−4.54%</b>   |
| <b>Operating expenses (before depreciation)</b> | <b>10.00%</b>            | <b>11.48%</b>             | <b>14.80%</b>   |
| Tax rate                                        | 30.00%                   | 40.14%                    | 33.80%          |
| Net working capital/Sales                       | 25.00%                   | 33.40%                    | 33.60%          |
| <b>Base-year sales for 2011 (\$000)</b>         | <b>\$1,000,000</b>       | <b>\$923,171</b>          | <b>−7.68%</b>   |

\*Critical value for each variable is the value of that variable that would, holding all other variables equal to their expected value, result in a zero NPV. This critical value is more descriptively called the breakeven value, because it produces a zero NPV. These estimates were found using the Goal Seek function in Excel.

To illustrate, consider the analysis presented in Table 2. The first column of the table identifies six key variables that are important for determining the NPV of the Earthlizer investment opportunity. The second column contains the expected values for each of the value drivers that we used in our analysis of the project's expected NPV. The third column contains the critical or breakeven values for each of the value drivers that results in a zero NPV for the project. The final column of the table compares the expected and critical values for these variables; it calculates the “% Change” in each variable from its expected value required to produce a zero NPV. This analysis suggests that three variables are particularly critical to the outcome of the Earthlizer investment, in that very slight deviations from their expected value have a significant impact on project NPV:

- gross profit margin (% change for breakeven = −4.54%)<sup>7</sup>
- operating expenses as a percent of sales (% change for breakeven = +14.80%)
- base-year sales for 2011 (% change for breakeven = −7.68%)

For example, if the gross profit margin declines to 31.12%, which is only 4.54% below the expected value of 32.60%, the project's NPV will drop to zero. Similarly, if operating expense as a percent of sales rises to 11.48% from the expected level of 10%, or if the initial year's sales are only \$923,171 rather than the projected \$1 million, then the NPV will drop to zero.

Knowing that these three value drivers are critical to the Earthlizer investment's success allows CSM's management to explore additional information regarding the likely values of each. Moreover, if the firm undertakes the investment, management can monitor each of these value drivers very closely, so as to be ready to take corrective actions quickly if the drivers deteriorate.

<sup>7</sup> Gross profit margin is technically an indicator of success or failure that is driven by two important value drivers: cost of goods sold and the markup the firm is able to charge over its costs. So when we analyze the gross profit margin as a determinant of value, we are implicitly evaluating the combined effects of the cost of goods sold (cost leadership) and the price markup over cost (product differentiation).



## TECHNICAL INSIGHT

### Using Microsoft Excel's “Goal Seek” Function

Goal Seek is a function within Excel (found under the “Tools” tab) that provides a powerful tool for performing “what if”-type analyses. For example, when you know the desired result of a single-cell formula (e.g.,  $NPV = 0$ ) but do not know the input value the formula needs to determine this result (e.g., revenue growth rate over the project’s life), you can use the Goal Seek feature to find the critical

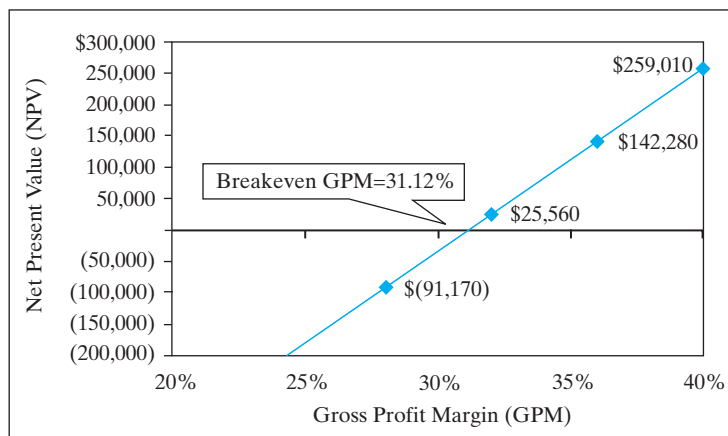
value of the input variable. In this example, Goal Seek iterates the value of the revenue growth rate until NPV is equal to zero.

The “Solver” tool in Excel is even more general than Goal Seek. Specifically, Solver allows us to solve maximization, minimization, or breakeven analysis problems in the presence of multiple constraints.

Figure 1 contains an NPV breakeven chart that illustrates the critical importance of gross profit margin to the success of the Earthilizer project. Our analysis assumes that the gross profit margin remains constant throughout the life of the project, and a gross profit margin of 31.12% in each year of the project’s life produces a zero NPV. Note that even very modest changes in the gross profit margin can have dramatic effects on the project’s NPV. For example if CSM, Inc. is successful in achieving a 36% gross profit margin (less than a 10.43% increase over the projected 32.6%), the estimated NPV for the project will more than triple in value to \$142,280. Unfortunately, similar changes in NPV occur if gross profit margins *drop* below the anticipated level. The clear message to the analyst is that gross profit margin is one of the most critical success factors for the project.

Although breakeven sensitivity analysis can be very helpful in identifying value drivers that are critical to project success, it has its limitations. First, this type of analysis

**Figure 1** NPV–Gross Profit Margin Breakeven Chart for the Earthilizer Investment Proposal (\$000)



considers only *one* value driver at a time, while holding all others equal to their expected values. This can produce misleading results if two or more of the critical value drivers are correlated with one another. For example, if the gross profit margin and the initial sales level both tend to be lower (or higher) than their respective expected values at the same time, then the breakeven sensitivity analysis will understate the true risks of the project.

Second, we don't have any idea about the probabilities associated with exceeding or dropping below the breakeven value drivers. For example, what is the probability that the gross profit margin will drop below 31.12% or that the initial sales level for the first year of the project will be \$923,171 or less? It would be very helpful to have some idea as to the likelihood of missing the key driver "targets."

Finally, we do not have a formal way of incorporating consideration for interrelationships among the variables. For example, gross profit margins and the operating expense to sales ratio are almost surely inversely related. It is important that we be able to consider this, and other interrelationships between the value drivers, in our analysis of the likely outcome from investing in the project.

To gain some insights into the distributions of possible valuation outcomes, we turn to simulation analysis, which offers analysts the ability to address all three of the above shortcomings of breakeven analysis.

## Simulation Analysis

**Monte Carlo simulation** provides a powerful tool that can help the analyst evaluate what *can happen* to an investment's future cash flows and summarize the possibilities in a probability distribution.<sup>8</sup> Simulation is particularly helpful in project analysis, because the outcomes from large investment projects are often the result of the interaction of a number of interrelated factors (or value drivers). This setting makes it very difficult to determine the probability distribution of a project's cash flows directly or analytically. However, we can simulate the distribution quite easily, as we illustrate in this section.

### Using Scenario Analysis As a Prelude to Building a Simulation Model

Scenario analysis and simulation can be complementary tools, because a scenario analysis can be an important prelude to the construction of a simulation model. Note that scenario analysis depends solely on the intuition of the decision maker; it helps the analyst think through the possible project outcomes and identify the key value drivers.

### Preparing and Running a Simulation

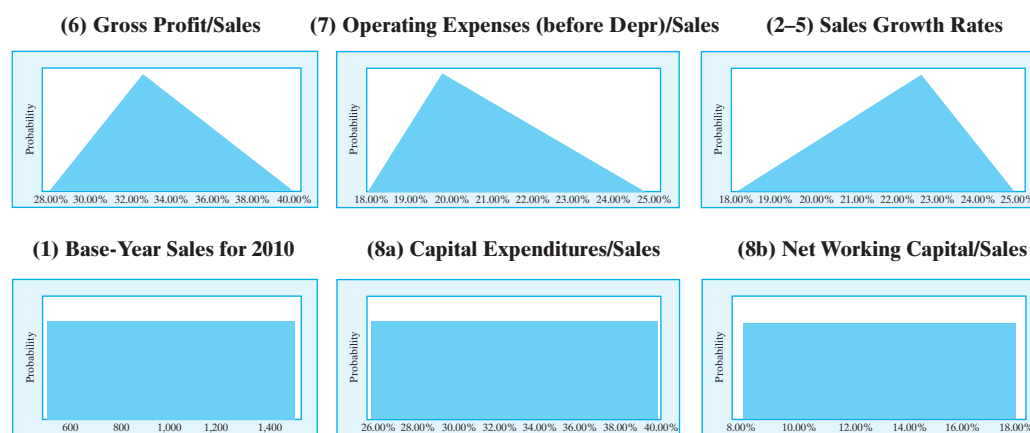
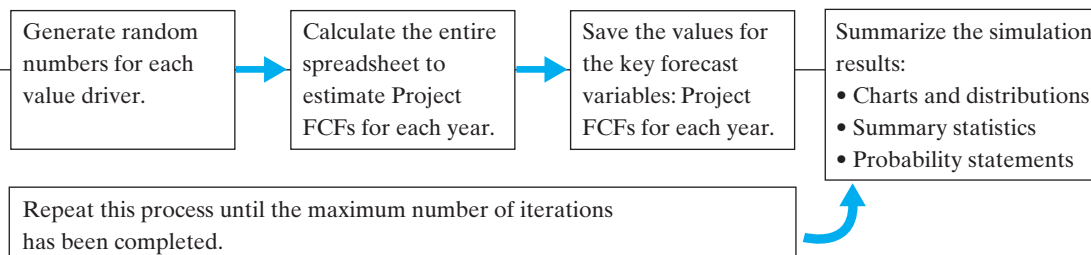
Figure 2 outlines the simulation process in three steps. In Step 1, we prepare a spreadsheet model that defines the investment's cash flows. In Step 2, we characterize each of the value drivers using a probability distribution. For example, we might ask the marketing department to describe its optimistic, most likely, and pessimistic estimates for the annual sales growth rate. The response might be that the highest annual growth rate (optimistic estimate) for the Earthilizer investment is 30% per year, the lowest possible (pessimistic estimate) growth rate is -10%, and the most likely growth rate is +10%. This "intuitive" information fully describes a triangular probability distribution (see the Technical Insight box on two popular probability distributions for simulation two pages ahead).

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<sup>8</sup> The term *Monte Carlo* as a form of simulation comes from the famous gambling casinos in Monaco. Gambling, in its pure form, is of course based on the rules of chance or probability, as is simulation.

**Figure 2** Steps in Running a Simulation Using the Earthilizer Example**Step 1: Prepare the spreadsheet model in Excel.**

|                                              | (\$000) | 2010               | 2011            | 2012            | 2013            | 2014             | 2015             |
|----------------------------------------------|---------|--------------------|-----------------|-----------------|-----------------|------------------|------------------|
| EBIT                                         |         | \$ —               | \$ 78.00        | \$ 100.60       | \$ 125.46       | \$ 152.81        | \$ 182.89        |
| Less: Taxes                                  |         | —                  | (23.40)         | (30.18)         | (37.64)         | (45.84)          | (54.87)          |
| NOPAT                                        |         | \$ —               | \$ 54.60        | \$ 70.42        | \$ 87.82        | \$ 106.96        | \$ 128.02        |
| Plus: Depreciation expense                   |         | —                  | 33.00           | 33.00           | 33.00           | 33.00            | 33.00            |
| Less: CAPEX                                  |         | (330.00)           | —               | —               | —               | —                | —                |
| Less: Changes in net working capital         |         | (250.00)           | —               | (25.00)         | (27.50)         | (30.25)          | (33.28)          |
| Plus: Liquidation of net working capital     |         |                    |                 |                 |                 |                  | 366.03           |
| Plus: Liquidation of PPE                     |         |                    |                 |                 |                 |                  | 165.00           |
| Equals: Project free cash flow (Project FCF) |         | <u>\$ (580.00)</u> | <u>\$ 87.60</u> | <u>\$ 78.42</u> | <u>\$ 93.32</u> | <u>\$ 109.71</u> | <u>\$ 658.77</u> |

**Step 2: Characterize the value drivers using a probability distribution.****Step 3: Run the simulation and interpret the results.**

To illustrate how this analysis might be carried out, in Table 3 we provide a set of hypothetical assumptions for the probability distributions used to describe the key value drivers in CSM's Earthilizer investment proposal. Specifically, there are eight value drivers that underlie the uncertainty in the investment outcome. These include (1) base-year sales, (2–5) annual rates of growth in revenues for 2012 through 2015 (four random

## TECHNICAL INSIGHT

### Two Popular Probability Distributions for Use in Simulation Models: The Uniform and Triangular Distributions

Among the set of distributions that can be used to model the uncertainty inherent in an investment, the uniform and triangular distributions are among the most popular. An important reason for this is that the variables that are needed to define the shape of both these distributions (known as their *parameters*) are very intuitive. Consequently, when the financial analyst is trying to elicit information about the proper distribution to use, he or she can ask straightforward questions of individuals who understand the underlying randomness of the variables that are being modeled.

A second reason for the popularity of these distributions relates to the fact that, although simple, they are very flexible and can be used to capture the essence of the randomness of lots of

random variables. This is particularly true of the triangular distribution, in that its form can be shaped and stretched (by adjusting the parameter values) to capture both symmetric and skewed distributions.

For example, in the uniform distribution, all values between the minimum and maximum are equally likely to occur. Consequently, the only things we need to know in order to completely define a uniform distribution (i.e., the parameters) are the minimum and maximum values that the random variable can take on. The mean of the uniform distribution is simply the sum of the maximum and minimum values divided by 2, because all values in the distribution have an equal probability of occurring. The upper figure

growth rates), (6) the gross profit margin, (7) operating expenses (before depreciation expense) to sales, and (8) the 2015 terminal value, which is a multiple of the book value of the capital invested in the project. Everything else in the model is assumed to be fixed.

We have purposefully limited ourselves to the use of the *triangular* and *uniform* distributions. The parameters of both distributions are very intuitive, thereby facilitating their estimation (see the Technical Insight box on choosing a probability distribution two

**Table 3** Monte Carlo Simulation Assumptions for the CSM Inc. Earthlizer Investment

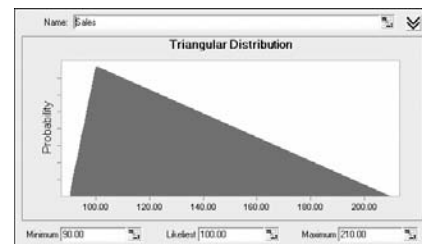
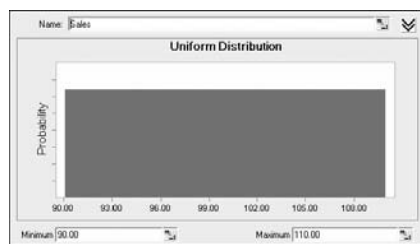
| Variable                                 | Expected Value | Distributional Assumption |                                               |
|------------------------------------------|----------------|---------------------------|-----------------------------------------------|
|                                          |                | Distribution              | Parameter Estimates                           |
| Base-year sales for 2010                 | \$1,000,000    | Uniform                   | Min = \$500,000 and Max = \$1,500,000         |
| Sales growth rates (2012–2015)*          | 10.0%          | Triangular                | Max = 30%, Most likely = 10%, and Min = –10%  |
| Gross profit margin = Gross profit/Sales | 32.6%          | Triangular                | Max = 40%, Most likely = 32.6%, and Min = 28% |
| Operating expenses (before depr.)/Sales  | 10.0%          | Triangular                | Max = 15%, Most likely = 10%, and Min = 5%    |
| Terminal value multiple of book value    | 1.00           | Uniform                   | Min = .5 and Max = 1.5                        |

\*There are actually four growth rates, one for each year, beginning with 2012 and ending with 2015.

on the following page depicts a uniform distribution for Sales, with a minimum value of \$90 and a maximum of \$110.

The triangular distribution, on the other hand, can be described by the minimum, maximum, and most likely values. The mean of the triangular distribution is equal to the sum of the minimum, most likely, and maximum values divided by 3. The lower figure depicts a triangular distribution for Sales where the minimum value is \$90, the most likely value is \$100, and the maximum value is \$210.

Note that the triangular distribution presented here is highly skewed toward larger values. However, the triangular distribution is very flexible and can be skewed in either direction and can also be symmetrical. For example, if you can imagine what would happen to the graph if you were able to grasp the peak of the triangle and stretch it to the right or left (within the minimum and maximum bounds of the distribution), you can envision the flexibility of the triangular distribution to reflect both skewed and symmetric probability distributions.



pages ahead). We use the triangular distribution to characterize the sales growth rate, gross profit margin, and operating expense to sales ratio. The parameters of the triangular distribution are the minimum, most likely, and maximum values for the underlying variable. We use the uniform distribution to characterize the base-year sales for 2011 and the terminal value multiple. The parameters of the uniform distribution are simply the minimum and maximum values.<sup>9</sup>

**Simulating project revenues** The annual cash flows of the Earthlizer investment are determined largely by the project's revenues, which we have modeled using two important drivers: the initial sales level for 2011 and the growth rates in revenue for 2012 through 2015. The initial sales level for 2011 has an expected value of \$1 million. However, because sales revenues for the project are subject to a high level of uncertainty, we assume that the sales level follows a uniform distribution with a minimum value of \$500,000 and a maximum value of \$1.5 million. These estimates are just that, estimates, and in most cases would come from the intuition and experience of the firm's marketing staff.

The sales level for 2012 is estimated using the simulated 2011 revenue number multiplied by 1 plus the rate of growth in revenues for 2012. We assume that the rate of growth in revenues for 2012 comes from a triangular distribution with a minimum value of -10%, a most likely value of 10%, and a maximum value of +30%. The dispersion of the minimum and maximum growth rates reflects the degree of uncertainty the analyst has about sales growth. Revenues for 2013–2015 are estimated in a similar fashion.

<sup>9</sup> Note that the parameter estimates for both the triangular and uniform distributions are consistent with the expected value used in the deterministic analysis of the investment. You can verify this by calculating the expected value for the triangular distribution as the sum of the minimum, most likely, and maximum values divided by 3. And for the uniform distribution, the mean is simply the sum of the minimum and maximum values divided by 2.

## TECHNICAL INSIGHT

### Choosing a Probability Distribution

When building a simulation model, it is necessary to identify not only the sources of randomness (uncertainty) that are the key value drivers in the model, but also an appropriate probability distribution to describe the randomness and estimate the parameters of that distribution. This is a daunting task, but there are some basic rules of thumb that can help the analyst:

- *First, if you have relevant data, then by all means use them.* Although we will not utilize the *distribution fitting* capability of Crystal Ball in this book, the more advanced user will find this an excellent way to take advantage of historical data related to the distributions being estimated.
- *Second, does the variable you are modeling assume only discrete values?* For example, if the variable you are estimating can only be zero or one (or yes or no), then a discrete distribution constructed using the custom category in Crystal Ball will work for you.
- *Third, select distributions that fit the wisdom of the experts from whom you will elicit parameter estimates.* For example, if your experts can tell you only the maximum and minimum values of a variable you are estimating, then the uniform distribution may be a reasonable choice, because it assigns the same probability to each value between a minimum and a maximum value. Alternatively, if your expert is willing to estimate a minimum and maximum as well as a most likely value, then the triangular distribution will be a good candidate for you. In this distribution, the probability associated with all values below the minimum or above the maximum value is zero, and the probabilities in between the maximum and minimum increase in a linear fashion from either extreme up to the most likely value (thereby forming a triangle).
- *Fourth, if there are theoretical reasons for selecting a particular distribution, then certainly do so.* For example, it is common practice to use a lognormal distribution to characterize market prices for financial securities.
- *Finally, the KISS principle (Keep It Simple, Stupid) is always appropriate.* Remember that you must elicit information from experts regarding the nature of the distribution to select and for parameter estimates. Because this may often require having discussions with individuals not well versed in the language of probability and probability distributions, it is probably best to err on the side of oversimplification rather than precision and sophistication.

**Forecasting project cash flows** Once we have forecasts of annual revenues, we calculate project cash flows by constructing a set of pro forma financial statements such as those presented earlier in Table 1. Note that the gross profit margin and operating expense to sales ratios are random variables (described in Table 3) that are determined in each of the iterations of the simulation. Based on the estimated revenues, the gross profit margin, and the operating expenses, we can calculate the project's pro forma income statement (found in Panel a of Table 1).

To complete our estimate of project free cash flow, we must estimate any new investment that might be needed for capital equipment (i.e., CAPEX) and net working capital. CSM's management estimates that, after the initial capital expenditure of \$330,000, the project will not require any more CAPEX over the period 2011–2015. However, as firm revenues change over time, they estimate that the project's need for net working capital will also rise and fall. Specifically, CSM estimates that the project



## T E C H N I C A L

## I N S I G H T

Simulation “Add-Ins”  
for Excel

There are two principal competing tools for adding easy-to-use simulation capabilities to your spreadsheet program. These include **@Risk** from the Palisade Corporation (<http://www.palisade.com/>) and **Crystal Ball Professional Edition** from Decisioneering Inc. (<http://www.crystalball.com/>). Both software packages offer similar capabilities; we have selected Crystal Ball for all our illustrations. If you are not familiar with Crystal Ball, you will find the short video presentation found on Decisioneering’s website very helpful in getting started.

requires net working capital equal to 25% of firm sales revenues. To accommodate this required investment, we include 25% of the revenues for 2011 as part of the initial investment outlay for 2010. In 2011, we adjust net working capital to equal 25% of realized (simulated) sales. Similarly, in 2012–2015 we set the project’s net working capital requirement equal to 25% of simulated sales for the year.<sup>10</sup>

Project free cash flows are then calculated following the method found in Panel c of Table 1. Specifically, for 2011 through 2014, we add depreciation expense back to net operating profit after taxes (NOPAT) and subtract the added investment in net working capital. Recall that CAPEX is assumed to be zero. The 2015 cash flow calculation involves one additional step—adding back the estimated terminal value of the Earthlizer project’s assets.

**Liquidating the investment and the terminal value** The cash flow in 2015 includes both an operating component (just like 2011–2014) plus a terminal value component. The latter deserves a bit of added explanation. We assume that in 2015 when the project is shut down, its accumulated working capital (current assets less current liabilities) plus the property, plant, and equipment can be sold for a multiple of their book value. This multiple is assumed to be a random variable that has a uniform distribution with a minimum value of .5 and a maximum value of 1.5 (this implies an expected value of 1.0). Note that if the terminal value multiple is equal to 1, there is no taxable gain or loss realized on the sale of the assets. However, in the event that the multiple is greater than 1, the sale creates a taxable gain that is taxed at the firm’s 30% tax rate. Should the assets be sold for a multiple less than 1, the resulting loss offsets taxable income in 2015 and results in a tax credit to CSM.

**Simulation software** The simulation analysis is carried out for the Earthlizer example using a spreadsheet “add-in” program known as **Crystal Ball Professional Edition**. (See the Technical Insight box above on simulation “add-ins” for Excel.) There are a

<sup>10</sup> We set the 2010 investment in net working capital equal to 25% of expected sales for 2011. In 2011, we simply adjust net working capital to reflect 25% of actual revenues. A more involved procedure would also accommodate an estimate of expected needs for net working capital for 2012. However, we have opted for simplicity in the model presented here and make the adjustments to net working capital at the end of the year when we know what annual sales for the period turned out to be. The effects of accommodating expected working capital needs have very little impact on the model outcome in this example.

number of other options available, including @Risk, as well as proprietary simulation software written for use in specialized applications of a given firm. A brief Appendix is found at the end of the chapter that summarizes the use of Crystal Ball Professional Edition for the beginner.

## Interpreting Simulation Results

The simulation uses 10,000 iterations to produce estimates of the distributions of Project FCFs for 2011–2015. To illustrate the nature of the simulation results, consider Panel a of Figure 3, which contains the distribution of simulated FCFs for 2011, as generated by Crystal Ball. The frequency distribution reflects 10,000 FCF values, characterized in terms of both the frequency with which they occurred (measured on the right-side vertical axis) and the corresponding probability of occurrence (measured on the left-side vertical axis).

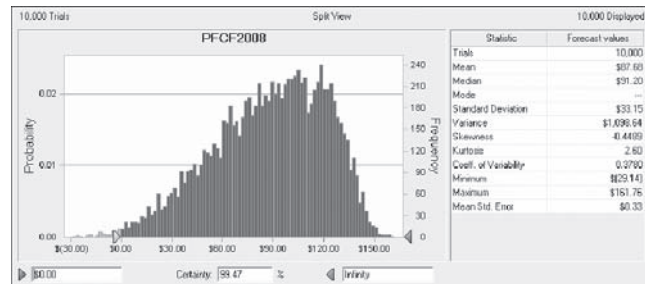
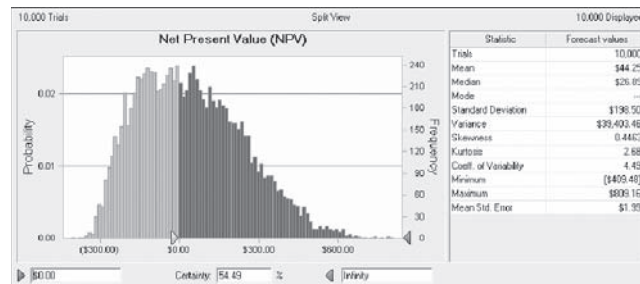
The simulated distribution of FCFs for 2011 (found in Panel a of Figure 3) provides some interesting insights into the prospects of the investment. First, there is substantial dispersion in the simulated cash flows for 2011, ranging from a minimum of \$29,140 up to a maximum of \$161,760, with a mean value of \$87,680. The “Certainty” cell found at the bottom of the graph indicates that 99.47% of the total area under the frequency distribution of FCFs lies above \$0.00 (i.e., the simulated FCFs are positive 99.47% of the time). Although we do not display them here, the FCF distributions for the remaining years of the project exhibit similar characteristics to those described above for 2011.

From the simulated cash flows in each year, we calculate the mean or average cash flow for each year from 2010 to 2015. These expected cash flows are found below (\$000):

|                                               | <b>2010</b>  | <b>2011</b> | <b>2012</b> | <b>2013</b> | <b>2014</b> | <b>2015</b> |
|-----------------------------------------------|--------------|-------------|-------------|-------------|-------------|-------------|
| Expected Project FCF<br>(from the simulation) | \$ (580,000) | 87,680      | 78,150      | 93,460      | 109,740     | 660,970     |
| Expected value of the<br>project              | \$624,218.13 |             |             |             |             |             |
| Expected NPV                                  | \$ 44,218.13 |             |             |             |             |             |
| Expected IRR                                  | 15.42%       |             |             |             |             |             |

Discounting the expected cash flows at the 13.25% cost of capital produces a \$624,218.13 present value, and subtracting the \$580,000 initial cost of the project in 2009 produces an expected NPV of \$44,218.13. Finally, based on the future Project FCFs and initial investment, we calculate an expected IRR of 15.42%. Note that the expected values of the NPV and IRRs calculated using the simulation differ slightly from our earlier estimates of NPV equal to \$43,070 and IRR equal to 15.36%. These differences reflect the fact that the simulated values converge to, but will not necessarily equal, their expected values.

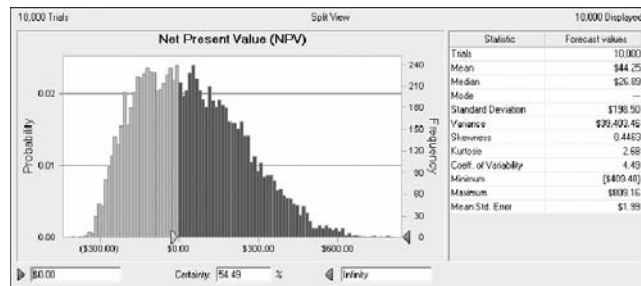
To this point, we have simply replicated our analysis of project valuation by simulating the distributions of cash flows that underlie the expected values reported earlier in Table 1. But this is just the beginning of our analysis. Let’s now look to see what we can learn about the project by analyzing the simulated distributions of NPV and IRR.

**Figure 3** Simulation Analysis of the Earthlizer Investment**Panel a. Distribution of Project FCFs for 2011 (\$000)****Panel b. Distributions of NPV and IRR from the Simulation****Analyzing the Distributions of NPV and IRR**

By simulating the distributions of NPV and IRR, we can calculate the probability that the investment will indeed realize an IRR that exceeds any given hurdle, like the firm's cost of capital or the risk-free rate. To estimate these values, we treat the yearly cash flows from 2011 to 2015 from each of the iterations in the simulation as a possible realization of project free cash flows and calculate the realized IRR and NPV given these cash flows.<sup>11</sup> Hence, if we do this simulation 10,000 times, we will have a distribution of 10,000 realized NPVs and IRRs.<sup>12</sup>

<sup>11</sup> The NPV and IRR for each of the iterations are part of the same simulation experiment that created the distributions of Project FCFs. We simply calculate a value for the NPV and IRR for each simulated set of Project FCFs.

<sup>12</sup> We need to be a bit careful here in the way that we interpret the distributions of NPV and IRR. The reason is that the mean of the distribution of NPV and IRR that we simulate and report in Panel b of Figure 3 is *not* equal to the NPV of the expected cash flows. The divergence in these numbers comes from something called Jensen's inequality. The issue of Jensen's inequality applies to nonlinear functions of random variables. Therefore, if the uncertainties arise in multiplicative form (for instance,  $\text{Sales}(t) = \text{Sales}(t-1)(1+g)$ , where both  $\text{Sales}(t-1)$  and growth are random variables—as in the case of the Earthlizer problem), Jensen's inequality will matter. Now Project FCF and NPV will be nonlinear functions of underlying random variables. The key thing to remember is that we have already estimated the expected NPV and IRR from the simulated distributions of cash flows for the project. From this point forward, we are simply investigating the underlying determinants of NPV and IRR (not their expected values).

**Figure 3** *continued***Legend:**

# *Trials*—the number of iterations in the simulation.

# *Displayed*—the number of iterations actually reflected in the frequency distribution. In the above analysis, all 10,000 iterations are included. However, the default in Crystal Ball is to omit outliers, such that the number of observations displayed is often less than the number of trials.

*Split View*—combines the frequency distribution with descriptive statistics. In addition to this view, a number of options are available, including cumulative frequency plots.

*Probability*—left-side vertical axis describes the simulated values in terms of their appearance as a fraction of the 10,000 iterations.

*Frequency*—right-hand vertical axis describes the simulated values in terms of the number of occurrences.

*Certainty*—displays the % of the area either above or below the critical value selected. In Panel a, the critical value is an FCF of \$0.00; in Panel b, it is an NPV of 0 and an IRR equal to the risk-free rate of 6%.

Panel b of Figure 3 presents the distributions of NPV and IRR generated from such a simulation. From these simulated values, we see that there is a 54.49% chance that the investment will generate cash flows that are sufficient to make the investment a positive-NPV investment at the firm's cost of capital (see the "Certainty" box at the bottom of the frequency distribution found in Panel b of Figure 3).<sup>13</sup> However, this also implies that there is a 45.51% chance that the realized NPV of the project will be negative. It is also the case that there is a 78.13% probability of realizing an IRR that exceeds the risk-free rate of 6%.<sup>14</sup> This indicates that the investment does have some risk, because there is a nontrivial probability that this risky investment will return less than the return on a perfectly safe investment like Treasury bonds.

What we have learned in the simulation does not provide a clear go–no go choice for CSM with regard to the Earthlizer project. However, we can further explore the sensitivity of the project NPV to each of the random variables in the model using a tool known as a *tornado diagram*.

<sup>13</sup> An observant reader will notice that the average IRR in this simulation is not equal to the IRR of the investment calculated using expected cash flows. The reason for this is the same as discussed in the previous footnote.

<sup>14</sup> We deliberately selected the risk-free rate as the "bar" over which we are asking the project to leap, because this analysis is designed to explore risk, not value the project. We did the latter when we calculated the expected NPV and expected IRR using the expected FCFs from the simulation.

## Using the Tornado Diagram to Perform Sensitivity Analysis

Earlier, we used breakeven sensitivity analysis to analyze the importance of the value drivers in a simple setting, where we analyzed possible changes in the value drivers one at a time. We now introduce a more powerful tool that combines scenario analysis and simulation to provide a systematic way to analyze how changes in each of the investment's value drivers impact the project NPV. Specifically, we can evaluate the sensitivity of the project valuation to variation in each of the value drivers using the **tornado diagram** found in Figure 4. The chart places the assumptions that have the greatest impact on project NPV at the top of the diagram and successively smaller ones down the figure, creating a funnel-shaped diagram—thus the name *tornado diagram*. (See the Technical Insight box on tornado diagrams, two pages ahead.)<sup>15</sup>

Here is how you interpret the numbers reflected in the tornado diagram. First, not surprisingly, the initial or base-year level of sales for 2011 has the greatest impact on the NPV of the Earthilizer investment. For example, if the 2011 sales level is set equal to its 90th percentile value of \$600,000 (the expected value is \$1 million), NPV falls to (\$181,160). However, if the 2011 sales level is equal to its 10th percentile value, which is \$1.4 million, the NPV rises to \$267,290. Clearly, the value of the investment is closely tied to the initial customer response to the product, which is critical to its overall success.

Behind the initial sales level in 2011, the next most important variable that drives the success or failure of the Earthilizer investment is the operating expense to sales ratio, followed by the gross profit margin and the terminal value multiple of the book value of invested assets. These results suggest the following responses:

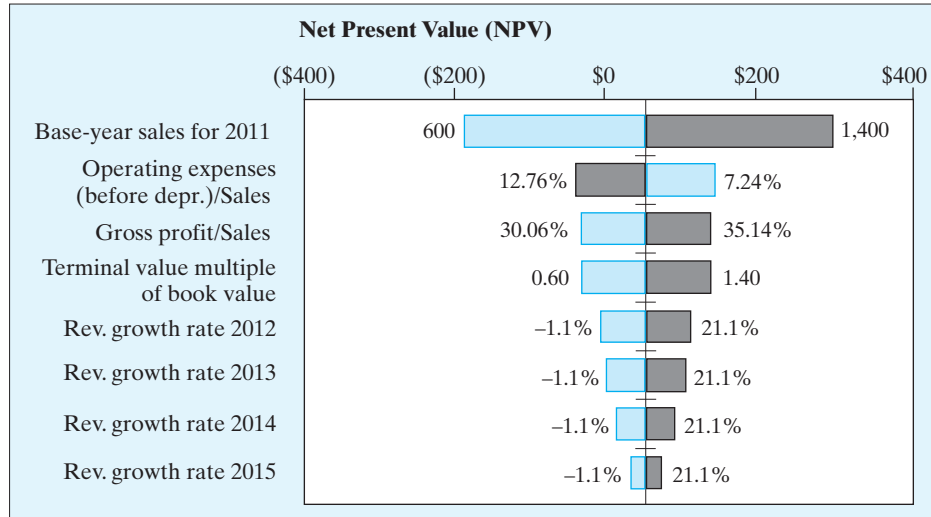
- First, to the extent that the viability of the investment is still in question, the analyst might invest additional time and money into learning more about these critical value drivers—specifically, the initial market response that drives 2011 sales. Might CSM take actions to learn more about the potential success of market entry (e.g., hold focus groups with potential customers or engage in pilot programs)? Are there things that CSM might do to ensure that initial sales (market penetration) are maximized? For example, promoting market awareness through an initial advertising campaign may be an effective tool for improving the chances of success of the project.
- Second, because operating costs are a critical value driver, management should pay particular attention to putting in place processes that can be used to monitor and improve operating efficiency.

## Summing Up the Simulation Results

So what have we learned from the simulation analysis about the Earthilizer investment? First and foremost, the project is likely to be a wealth-enhancing venture (i.e., it has a positive NPV). However, the outcome is far from certain. In fact, based on our analysis, it appears that there is a 78.13% chance that the project will return more than the 6% rate of return on risk-free Treasury bonds (and a 21.87% chance that it will earn less). In addition, we learned that the key value drivers are the initial sales level for 2010, the operating expenses to sales ratio, the terminal valuation multiple for the working capital

<sup>15</sup> The tornado diagram provides a “one-at-a-time” form of sensitivity analysis. Crystal Ball also provides a sensitivity analysis tool that uses the simulated values to analyze the rank correlation between forecast variables and each of the assumptions.

**Figure 4** Tornado Diagram for the Earthlizer Investment Proposal (NPV \$000)



| Variable                                | Net Present Value (NPV) |           |          | Input    |        |           |
|-----------------------------------------|-------------------------|-----------|----------|----------|--------|-----------|
|                                         | Downside                | Upside    | Range    | Downside | Upside | Base Case |
| Base-year sales for 2011                | (\$181.16)              | \$267.29  | \$448.44 | 600      | 1,400  | 1,000     |
| Operating expenses (before depr.)/Sales | \$123.72                | (\$37.59) | \$161.31 | 12.76%   | 7.24%  | 10.00%    |
| Gross profit/Sales                      | (\$31.14)               | \$117.27  | \$148.41 | 30.06%   | 35.14% | 32.60%    |
| Terminal value multiple of book value   | (\$29.60)               | \$115.74  | \$145.34 | 0.60     | 1.40   | 1.00      |
| Rev. growth rate 2012                   | (\$1.83)                | \$87.96   | \$89.79  | -1.1%    | 21.1%  | 10.0%     |
| Rev. growth rate 2013                   | \$9.29                  | \$76.85   | \$67.56  | -1.1%    | 21.1%  | 10.0%     |
| Rev. growth rate 2014                   | \$20.08                 | \$66.05   | \$45.97  | -1.1%    | 21.1%  | 10.0%     |
| Rev. growth rate 2015                   | \$30.57                 | \$55.56   | \$24.99  | -1.1%    | 21.1%  | 10.0%     |

**Legend:**

*Tornado chart*—measures the input of each variable one at a time, independently, on the target forecast (NPV in this example).

*Range bars*—illustrate the impact of each of the input variables (Gross profit/Sales, and so forth) on NPV when the input variables are set equal to their 10th and 90th percentile values. For example, if the Gross profit/Sales ratio drops to its 10th percentile value of 30.06%, then NPV is equal to (\$31,140). Alternatively, if the Gross profit/Sales ratio is equal to its 90th percentile value of 35.14%, then NPV rises to \$148,410. The color of the bar indicates whether the change in the input is positive or negative. For example, an increase in the Gross profit/Sales ratio produces a positive effect on NPV, whereas an increase in the Operating expenses (before depr.)/Sales ratio results in a decline in NPV. The order of the input variables in the chart ranks them from the greatest to the least impact on NPV.

## TECHNICAL INSIGHT

### Tornado Diagrams

The **tornado diagram** can be used to evaluate each assumption's relationship to the forecast variables. In Figure 4, the assumptions are set equal to, say, their 10th and 90th percentile values and these extreme values are used to calculate the forecast variable of interest (e.g., project NPV or IRR). Note that the value of the forecast variable is read along the horizontal axis at the top of the diagram, and the 10th and 90th percentile values of the assumption are printed at the end of the bar that identifies their impact on the forecast variable. The assumptions that have the greatest impact on the forecast variable are placed at the top of the diagram, followed by the assumptions that have the next largest impact.

and fixed assets that are sold at the end of the project's life, and the gross profit margin. Consequently, a key determinant of the success or failure of the investment will be the ability of the firm's management to market the product aggressively from the outset and contain the manufacturing costs underlying the cost of goods sold and the operating costs associated with running the business.

Ultimately, however, the decision to undertake the investment is a matter of judgment, and it will depend on the availability of other projects, the amount of capital at risk, and the individuals involved. Simulation analysis is simply another tool for adding information about the nature of the uncertainty underlying the project and its prospects for success. It should be viewed as a decision tool, not a decision maker that provides clear yes/no criteria.

### Reflections on the Use of Simulation

Building a simulation model requires the analyst to think deeply about the underlying sources of uncertainty that affect an investment's profits and forces him or her to be very explicit about assumptions that might otherwise be implicit. Consequently, there are benefits to simulation analysis that arise from the process itself as well as from the actual output of the simulation.

For example, an analyst could forecast the cash flows generated by the Earthlizer investment without the benefit of a simulation model. These estimates would rely upon estimates of the expected values of each of the value drivers (i.e., sales growth rates, gross profit margins, operating expense ratios, and so forth). However, without the need to model the distributions of the value drivers, the analyst might simply rely upon the "best guess" for their expected values. Such "from-your-gut" forecasting may work reasonably well for very simple investments, but it is problematic for more complex cases in which there are multiple sources of uncertainty that interact with one another to determine the distribution of project cash flows.

It is sometimes argued that the difficulty encountered in making these distributional assumptions is reason enough not to do them. However, the fact is that the analyst who fails to address the underlying complexities of an investment's cash flows explicitly is simply making the assumptions implicit in the analysis. As a general rule, we think that it is better to be as explicit as possible when evaluating major investments.



## PRACTITIONER INSIGHT

### Sensitivity Analysis at ConocoPhillips: A Conversation with Steven McColl\*

Sensitivity analysis has been a long-established technique in the oil and gas industry when analyzing multimillion-dollar investments. The construction of simulation models to evaluate investments includes the formation of multidisciplinary teams of engineers, commercial specialists, and finance experts to identify and quantify the risks and uncertainties present in each investment.

ConocoPhillips, like many in its peer group, has developed a process for the evaluation of risks and uncertainties to ensure the validity of the numbers entering the financial models. The overall evaluation process includes:

- Establishing a group of experts to understand the problem framework
- Developing technical and financial models to analyze risk and uncertainties

- Providing insight and understanding from those models to management in order to maximize profit and mitigate risks

Investment analysts at ConocoPhillips use simulation to calculate expected cash flows by incorporating uncertainty ranges for oil and gas prices, capital expenditures, operating costs, the amount of reserves, production rates, and tax regimes. The outputs from these models are expressed as Expected Monetary Value (EMV), IRR, Profitability Index, Period to Payback, and a number of other company- and industry-specific measures that describe the risk and reward from the project.

\*Strategic transactions coordinator, ConocoPhillips, Aberdeen, Scotland.

### Extensions of the Earthlizer Model

There are virtually limitless possibilities for refining and modifying the simulation model we used to analyze the Earthlizer investment proposal. An important opportunity for improving our model relates to incorporating consideration for dependence into the sources of randomness that drive the project cash flows. Specifically, there are two forms of dependence that are important in modeling project cash flows. The first relates to dependence that evidences itself over time. For example, if project revenues in 2011 are higher than expected, this may suggest that the project will grow more rapidly than if the initial year's revenue is lower than expected.

The second form of dependence relates to the possibility that within any given year, some of the expenses may vary together. For example, cost of goods sold (which is captured in the gross profit margin) and operating expenses as a percent of firm sales may move together. This would be the case if the factors that impinge on the firm's ability to operate efficiently and at relatively low cost (labor cost and so forth) are also linked to the factors that drive the firm's costs of goods sold.

If dependencies of the types described above are important, then the simulation can be modeled in such a way as to capture them. Although we will not dig deeper into the techniques that might be used, suffice it to say that the analyst can model the dependence directly either by linking the values of one value driver to another or by incorporating correlations into the process using Crystal Ball.

The final modeling extension we mention relates to the notion of **mean reversion** in year-to-year revenue growth rates. By this we refer to the fact that it is quite common

for growth rates in revenues and earnings to demonstrate the following behavior. If the growth rate in revenues for 2012 is above the expected rate of growth for that year, the rate of growth in revenues for 2013 is likely to be lower than expected. This particular pattern of growth rates can be induced into the simulation model by incorporating negative correlations between the adjacent year's rates of growth in revenue.

#### Using Simulations to Estimate an Investment's Contribution to Firm and Market Risk

Simulation provides a powerful tool for learning about the value-creation potential of a proposed investment. Specifically, it provides a method for simultaneously incorporating multiple and interacting sources of uncertainty into the analysis of the distribution of an investment's cash flows. Consequently, we can use simulation to solve what would otherwise be very complex estimation problems. We should note, however, that in our example, as well as in most industry applications, project risks are considered in isolation.

To get a more complete picture of an investment's risk, one should conduct simulations that examine how the risk of the project contributes to the overall risk of the firm, and perhaps also the aggregate economy.<sup>16</sup> For example, one might consider the realization of the firm's overall cash flows in the different scenarios and then use simulations to determine the covariance between the proposed investment cash flows and the overall firm cash flows. An investment with a lower covariance would be deemed less risky, because it helps diversify the firm.

## 4 DECISION TREES—VALUING PROJECT FLEXIBILITY

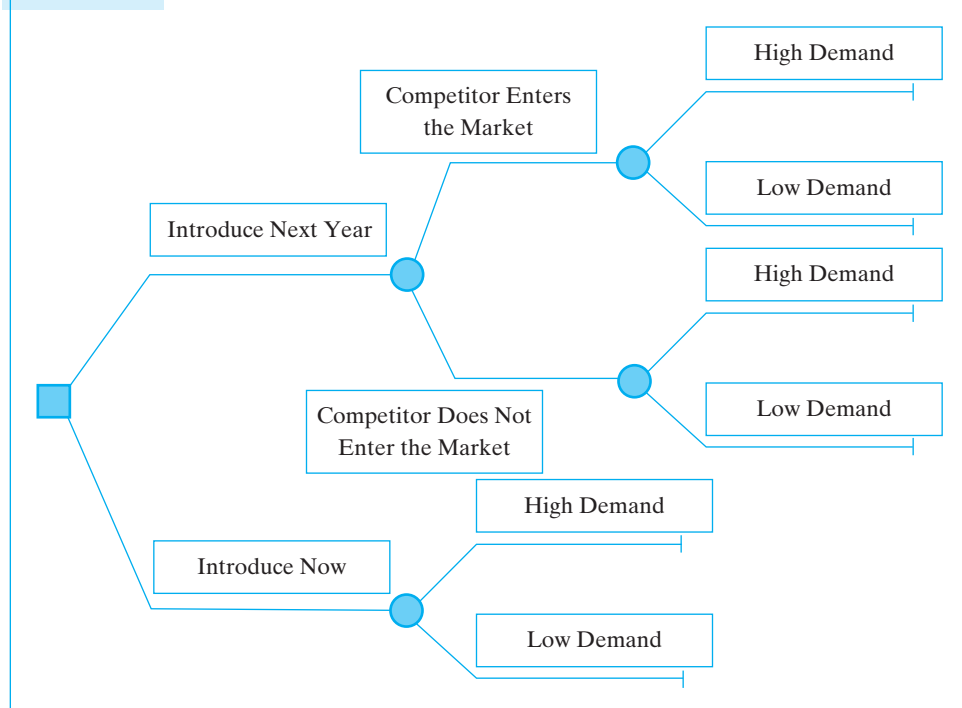
Up to this point, we have assumed that the firm has a set way to implement the investment project and will not deviate from this plan. While this greatly simplifies the analysis, in reality, most investment projects offer the firm some flexibility as to the inputs used or even the final products produced. For example, rather than proceeding with a fixed production schedule, firms tend to expand capacity and thus generate greater cash flows when market demand is higher, and similarly, they may cut back on capacity when market demand is lower.

The astute student of finance will recognize that we are talking about *options* associated with investments. For now, we will introduce the topic so that readers can recognize how flexibility in the implementation of an investment affects the expected cash flows that are used in a DCF analysis.

A decision tree is a very useful tool that illustrates the degree of flexibility in a project's implementation and how future decisions can affect values. For example, the decision tree found in Figure 5 describes the consequences of a decision to introduce a new product next year versus now. If the firm delays the product introduction, there is the possibility that a competitor may enter, whereas offering the product immediately reduces the likelihood of competition but also reduces the chances of a successful launch.

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<sup>16</sup> By examining how the project cash flows correlate with the aggregate economy, we can indirectly gauge the systematic risk of the project, which is a determinant of its required rate of return.

**Figure 5****Decision Tree for Timing a New Product Offering**

Note that the decision tree contains a number of nodes identified by vertical lines, circles, and boxes. The vertical lines denote terminal nodes that signal the ends of decision processes. The circles signify event nodes that represent points where nature intervenes and something happens that is subject to chance. In this decision tree, there are only two possible outcomes (High Demand and Low Demand). Finally, the square at the origin of the decision tree depicts a decision node that represents a point where the decision maker determines what happens. For example, will the new product be introduced immediately or delayed for one year?

Although we do not include additional decision nodes, it is conceivable that there might be many opportunities to make decisions over the life of the investment. For example, if the product is introduced now and the firm faces low demand, it might then face the decision to either continue marketing the product or abandon the investment altogether.

### Example—Decision Tree Analysis of the Abandonment Option

To illustrate how we can use decision trees to incorporate flexibility considerations into the analysis of investment projects, consider the following “modification” of the Earthilizer example. Assume that CSM Inc. faces an EPA test of the runoff effects of Earthilizer on groundwater. If, at the end of one year, the EPA determines that the project poses no hazard, it will give Earthilizer its approval for continued sale in its current form. There will be no effect on expected project cash flows.

If, on the other hand, the EPA finds that Earthlizer has a detrimental effect on groundwater, then further processing will be required that will cost CSM an estimated \$80,000 per year in processing costs (after taxes). Based on its own studies, CSM's management "guesstimates" that there is a 20% chance that the EPA will force the company to spend the additional funds.

### Evaluating the Revised Earthlizer Project Without the Abandonment Option

Let's evaluate the project cash flows and corresponding NPV for the Earthlizer investment, revised to take into account the EPA test. To illustrate the importance of the option to abandon after one year if the EPA ruling goes against the firm, we will first evaluate the revised cash flows assuming that the firm continues to operate even if it incurs the EPA costs.

In this case, the expected cash flows are equal to a weighted average of the original cash flows from Panel c of Table 1 and the revised cash flows. The latter amount is simply the original expected cash flows less the \$80,000 in additional processing costs. The weights we apply to the two cash flows equal the probability that the EPA ruling will be favorable and the probability that it will go against the firm (i.e., 1 minus the probability of a favorable outcome). Consequently, for 2011, the revised estimate of the expected FCF is calculated as follows:

$$\text{Revised FCF}_{2010} = .8 \times \$87,600 + (1 - .8)(\$87,600 - 80,000) = \$71,600$$

Panel a of Table 4 presents the complete set of revised FCFs for each year. The expected NPV for the Earthlizer project using these revised cash flows is a negative (\$12,872). In other words, the risk of having to incur \$80,000 in added costs makes the NPV of the project negative. This suggests that the project should not be undertaken at all. Note, however, that we are assuming that the project will be continued for the complete five-year planning period regardless of what the EPA ruling turns out to be. Let's see the effect of the option to abandon.

### The Value of the Abandonment Option

Panel a of Figure 6 describes the situation faced by Earthlizer: The firm has the option to abandon after one year should an unfavorable EPA ruling require that the firm spend an additional \$80,000 per year (after taxes). If CSM decides to shut down its Earthlizer operations, it anticipates that it can recoup \$380,000 of its \$580,000 initial investment, plus the first year's expected project free cash flow of \$87,600. The character of the investment in Earthlizer changes substantially when we add this additional source of uncertainty to the problem, in conjunction with the potential for shutting down operations and recouping most of the firm's initial investment.

In Panel b of Figure 6, we evaluate the NPVs of each of the four possible circumstances the firm might face after one year:

- If the EPA ruling is favorable, then operating the Earthlizer project over its full five-year expected life has an NPV of \$43,062.
- Moreover, if it abandons the project after one year, it will experience a negative expected NPV of (\$167,108).<sup>17</sup>

<sup>17</sup> Note that this is the NPV of abandoning at the end of Year 1 whether the EPA rules favorably or unfavorably.

**Table 4** Revised Cash Flows and NPVs for the Earthilizer Project

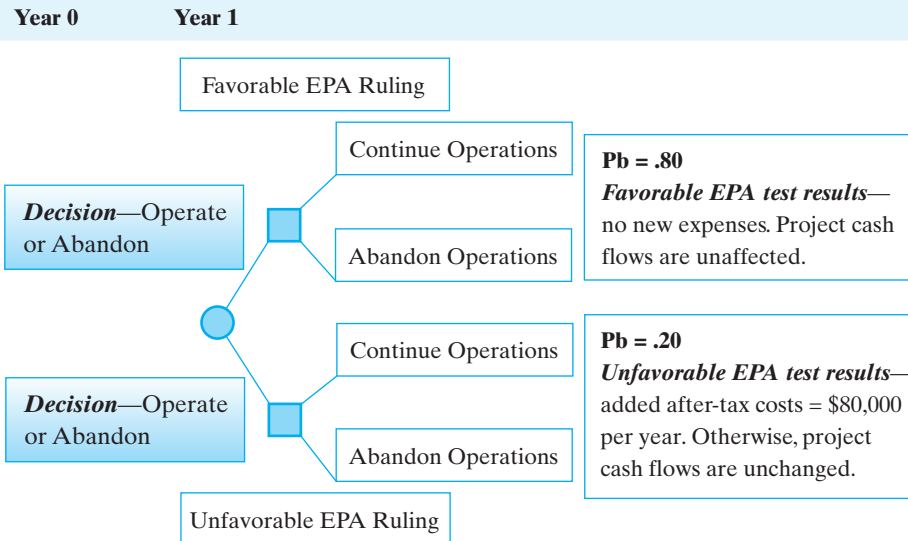
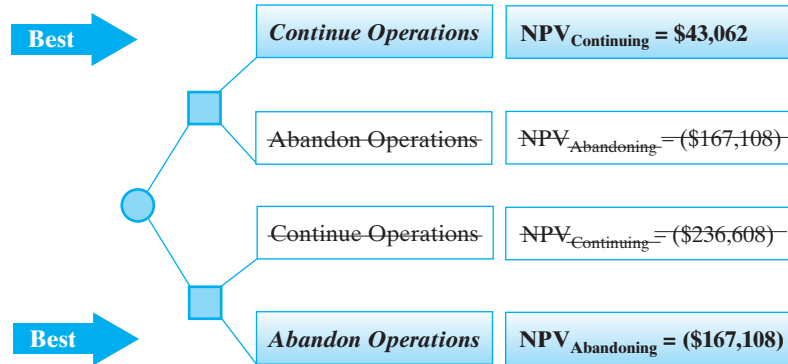
| <b>Panel a. No Option to Abandon</b>         |              |             |             |             |             |             |
|----------------------------------------------|--------------|-------------|-------------|-------------|-------------|-------------|
|                                              | <b>2010</b>  | <b>2011</b> | <b>2012</b> | <b>2013</b> | <b>2014</b> | <b>2015</b> |
| Favorable EPA ruling—                        |              |             |             |             |             |             |
| Expected FCFs                                | \$ (580,000) | \$87,600    | \$ 78,420   | \$93,320    | \$109,710   | \$658,770   |
| NPV (favorable EPA ruling)                   | \$ 43,062    |             |             |             |             |             |
| Unfavorable EPA ruling—                      |              |             |             |             |             |             |
| Expected FCFs                                | \$ (580,000) | \$ 7,600    | \$ (1,580)  | \$13,320    | \$ 29,710   | \$578,770   |
| NPV (unfavorable EPA ruling)                 | \$ (236,608) |             |             |             |             |             |
| Revised expected FCFs                        | (580,000)    | 71,600      | 62,420      | 77,320      | 93,710      | 642,770     |
| E[NPV] with no option to abandon             | \$ (12,872)  |             |             |             |             |             |
| <b>Panel b. Option to Abandon</b>            |              |             |             |             |             |             |
|                                              | <b>2010</b>  | <b>2011</b> | <b>2012</b> | <b>2013</b> | <b>2014</b> | <b>2015</b> |
| Project not abandoned (favorable EPA ruling) | (580,000)    | 87,600      | 78,420      | 93,320      | 109,710     | 658,770     |
| NPV (favorable EPA ruling)                   | 43,062       |             |             |             |             |             |
| Project abandoned (unfavorable EPA ruling)   | (580,000)    | 467,600     | —           | —           | —           | —           |
| NPV (unfavorable EPA ruling)                 | \$(167,108)  |             |             |             |             |             |
| E[NPV] with abandonment option               | \$ 1,028.31  |             |             |             |             |             |

Obviously, given these two choices, if the EPA ruling is favorable, CSM will want to operate the project. This alternative is highlighted, and the inferior alternative (abandoning) is struck out.

What if the EPA ruling is unfavorable? Again, Panel b of Table 4 presents the numbers:

- If the EPA ruling is unfavorable, the expected NPV of continuing operations is (\$236,608).
- The NPV of abandoning the project is only (\$167,108).

So, in the event of an unfavorable ruling, CSM should abandon the project. Once again, the superior alternative is highlighted, while the inferior alternative is struck out.

**Figure 6** Decision Tree—Abandonment Option**Panel a. Decision Tree for the Earthlizer Investment****Panel b. Project NPVs for Decision Tree Branches**

## Project Risk Analysis

If the probability of a favorable ruling is 50%, the expected NPV for the project in the presence of the option to abandon is now:

$$\begin{aligned} \text{Expected Net Present Value} &= \left[ \begin{array}{c} \text{Expected Net Present Value} \\ \text{Continue Operations – Favorable} \\ \text{EPA Response} \end{array} \right] \times \left[ \begin{array}{c} \text{Probability} \\ \text{of a Favorable} \\ \text{Ruling} \end{array} \right] \\ \text{(with the abandonment option)} &+ \left[ \begin{array}{c} \text{Expected Net Present Value} \\ \text{Discontinue Operations –} \\ \text{Unfavorable EPA Response} \end{array} \right] \times \left[ \begin{array}{c} \text{Probability of} \\ 1 - \text{a Favorable} \\ \text{Ruling} \end{array} \right] \quad (1) \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Expected Net Present Value} &= (\$43,062 \times .80) + [-\$167,108 \times (1 - .80)] = \$1,028 \\ \text{(with the abandonment option)} \end{aligned}$$

In the event of a negative ruling from the EPA, the option to shut down or abandon the Earthilizer project at the end of Year 1 makes the expected NPV of the project positive. The key here is that the firm can *abandon the project* in order to avoid the losses incurred by operating the project.

Abandonment truncates the lower tail of the distribution of a project's NPV. The truncation eliminates negative-NPV outcomes and, as a result, increases the mean of the now-truncated distribution to \$1,028. Without the option to abandon, the expected NPV was negative and equal to -\$12,872. Essentially, the abandonment option mitigates some of the risk of an unfavorable EPA ruling. Clearly, considering the effects of project flexibility (in this case, the flexibility to reconsider the investment and possibly abandon it should the EPA ruling prove unfavorable) in the analysis of the project has a material effect on project valuation.

### Using Option Pricing Theory to Evaluate the Abandonment Option

In our current example, we simply assume that we knew what the project would be worth both with and without a favorable response from the EPA.

## 5 SUMMARY

Determining the value of an investment opportunity is straightforward when future cash flows are known. However, we live in a very uncertain world, and very few investments generate cash flows that can be predicted with any degree of precision. While uncertainty clearly complicates the valuation process, a number of tools help the analyst deal with this complexity more effectively.



This chapter introduced three basic tools that are used to assess the impact of uncertainty on project outcomes: scenario analysis, breakeven sensitivity analysis, and simulation analysis. All three tools provide information that the analyst can use to understand the key factors that drive a project's success or failure. This information helps the analyst develop a better understanding of how confident the firm should be in the project's prospects.

This chapter also introduced the important role played by *project flexibility*. By this we mean the opportunities for the firm to modify an investment over its life in response to changing circumstances and new opportunities. It is useful to begin thinking now about how flexibility can affect expected cash flows as well as cash flow risks.

## PROBLEMS

**1 BREAKEVEN SENSITIVITY ANALYSIS** The Clayton Manufacturing Company is considering an investment in a new automated inventory system for its warehouse that will provide cash savings to the firm over the next five years. The firm's CFO anticipates additional earnings before interest, taxes, depreciation, and amortization (EBITDA)<sup>18</sup> from cost savings equal to \$200,000 for the first year of operation of the center; over the next four years, the firm estimates that this amount will grow at a rate of 5% per year. The system will require an initial investment of \$800,000 that will be depreciated over a five-year period using straight-line depreciation of \$160,000 per year and a zero estimated salvage value.

- a. Calculate the project's annual free cash flow (FCF) for each of the next five years, where the firm's tax rate is 35%.
- b. If the cost of capital for the project is 12%, what is the projected NPV for the investment?
- c. What is the minimum Year 1 dollar savings (i.e., EBITDA) required to produce a breakeven NPV = 0?

**2 PROJECT RISK ANALYSIS—SENSITIVITY ANALYSIS** Answer the following questions:

- a. What are the key sources of risk that you see in this project?
- b. Use the "Goal Seek" function within Excel to find the breakeven values (i.e., values that force the project NPV to equal zero) for each of the following variables: the initial CAPEX, the working capital percentage of revenue growth, variable cost % of sales, and sales volume. (*Hint:* Scale the sales volume for all five years up and down by the same percentage.)
- c. Which of the variables analyzed in part b do you think is the greatest source of concern? What, if anything, could you do to reduce the risk of the project?
- d. Should you always seek to reduce project risk?

<sup>18</sup> EBITDA is a widely used measure of firm earnings that we will encounter many times. It is simply earnings before interest and taxes, or EBIT, plus depreciation and amortization expense.

**3 PROJECT RISK ANALYSIS—COMPREHENSIVE** Bridgeway Pharmaceuticals manufactures and sells generic over-the-counter medications in plants located throughout the Western Hemisphere. One of its plants is trying to decide whether to automate a portion of its packaging process by purchasing an automated waste disposal and recycling machine.

The proposed investment is \$400,000 to purchase the necessary equipment and get it into place. The machine will have a five-year anticipated life and will be depreciated at a rate of \$80,000 per year, toward a zero anticipated salvage value. The firm's analysts estimate that the purchase of the new waste-handling system will bring annual cost savings of \$40,000 from reduced labor costs, \$18,000 per year from reduced waste disposal costs, and \$200,000 per year from the sale of reclaimed plastic waste net of selling expenses. Bridgeway requires a 20% return from capital investments and faces a 35% tax rate.

- a. Using the estimates provided above, should Bridgeway purchase the new automated waste-handling system?
- b. The manager at the plant where the handling system is being contemplated has raised some questions regarding the potential savings from the system. He asked the financial analyst in charge of preparing the proposal to evaluate the impact of variations in the price of plastic waste materials, which have proven to be volatile in the past. Specifically, what would be the impact of price reductions for the waste that drive the revenues from the sale of waste down to half their estimated amounts in Years 1 through 5?
- c. (*Simulation*) Model the new investment, whose value is determined by the following random variables: Annual revenues from reclaimed waste in Year 1 follow a triangular distribution with a minimum value of \$100,000, a most likely value of \$200,000, and a maximum value of \$300,000. In Year 2 (and each year thereafter), the distribution is still triangular; however, the most likely value is now equal to the value observed in the previous year. The minimum value is equal to 50% of the observed value in the previous year, and the maximum is equal to 150% of the observed value in the previous year. Furthermore, the revenues from reclaimed waste exhibit a correlation coefficient from year to year of .90. Labor cost savings can be forecast with a high degree of certainty, because they represent the savings from one hourly worker that will no longer be needed once the new waste-handling system has been put into place. The reductions in waste disposal costs come from a uniform distribution, with minimum value of \$15,000 and maximum value of \$21,000. The waste disposal costs are assumed to be uncorrelated over time.
  - i. What is the probability of a cash flow less than \$150,000 in Year 1? In Year 5? (*Hint:* Define the annual Project FCFs for Years 1 through 5 as *forecast* variables. You will use only the Years 1 and 5 cash flow distributions for this question but will use all of them to answer part iii.)
  - ii. What are the expected NPV and IRR for the project?
  - iii. (Optional) What are the means and standard deviations of the simulated distribution of cash flows for Years 1 through 5? What is the effect of the positive correlation in the underlying determinants of the project's cash flows? (*Hint:* Look at the standard deviations in annual cash flows through time.)

**4 PROJECT RISK ANALYSIS—BREAKEVEN SENSITIVITY** The TitMar Motor Company is considering the production of a new personal transportation vehicle that would be called the PTV. The PTV would compete directly with the innovative new Segway.

The PTV will utilize a three-wheel platform capable of carrying one rider for up to six hours per battery charge, thanks to a new battery system developed by TitMar. TitMar's PTV will sell for substantially less than the Segway but will offer equivalent features.

The pro forma financials for the proposed PTV project, including the forecasts and assumptions that underlie them, are set out in Exhibit P4.1. Note that revenue is calculated as follows: price per unit  $\times$  market share (%)  $\times$  market size and units sold = revenues/price per unit. The project offers an expected NPV of \$9,526,209 and an IRR of 39.82%. Given TitMar's stated hurdle rate of 18%, the project looks like a winner.

Even though the project looks very good based on management's estimates, it is risky and can turn from a positive-NPV investment to a negative one with relatively modest changes in the key value drivers. Develop a spreadsheet model of the project valuation and answer the following questions.

- a. If the firm's market share turns out to be only 5%, what happens to the project's NPV and IRR?
- b. If the market share remains at 15% and the price of the PTV falls to \$4,500, what is the resulting NPV?

**5 SIMULATION ANALYSIS** Use your model from Problem 4 to construct a simulation model for the TitMar PTV. Incorporate two random (i.e., stochastic) variables in your model to capture the size of the market for scooters. Each of these variables is modeled differently as follows:

**MARKET SHARE.** Market share follows a triangular distribution with a most likely value of 15%, a minimum of 10%, and a maximum of 20%.

**GROWTH RATE IN MARKET SIZE.** The growth rate in market size for Year 1 is assumed to be normally distributed with a mean of 5% and standard deviation of 2%. For Year 2, the expected market-size growth rate is equal to the simulated growth rate for Year 1 and has a standard deviation of 2%. The growth rates for Years 3 and beyond follow the pattern described for Year 2.

Run 10,000 random trials and define two output variables—net present value and internal rate of return. What is the probability that the NPV will be zero or lower? What is the probability of an IRR less than 18%?

**6 DECISION TREE AND BREAKEVEN SENSITIVITY ANALYSIS** Reevaluate the Earthlizer decision tree analysis found in Table 4, where the abandonment value is only \$350,000. What is the expected NPV of the project under these circumstances? What is the minimum abandonment value required to produce an expected NPV for the project of zero with the abandonment option? (*Hint:* Use the “Goal Seek” function to solve for the abandonment value that produces an expected NPV of zero.)

**7 INTRODUCTORY SIMULATION EXERCISES** Construct a spreadsheet model for each of the following exercises and then use the model to build a simulation model.

- a. Jason Enterprises faces uncertain future sales. Specifically, for the coming year, the firm's CFO described her sales expectations as follows: “Sales could be as high as \$10,000,000 or as low as \$7,000,000, but I could not tell you anything more.” How would you characterize firm sales using a probability distribution? If Jason's operating earnings are typically 25% of firm sales, what would you estimate earnings

**Exhibit P4.1****TitMar Motor Company Personal Transportation Vehicle (PTV) Project**

| <b>Assumptions and Predictions</b>             |               | <b>Estimates</b>  |                                                |               |               |               |  |
|------------------------------------------------|---------------|-------------------|------------------------------------------------|---------------|---------------|---------------|--|
| Price per unit                                 |               | \$                | 4,895                                          |               |               |               |  |
| Market share (%)                               |               |                   | 15.00 %                                        |               |               |               |  |
| Market size (Year 1)                           |               |                   | 200,000 units                                  |               |               |               |  |
| Growth rate in market size beginning in Year 2 |               |                   | 5.0 %                                          |               |               |               |  |
| Unit variable cost                             |               | \$                | 4,250                                          |               |               |               |  |
| Fixed cost                                     |               |                   | \$9,000,000                                    |               |               |               |  |
| Tax rate                                       |               |                   | 50.0 %                                         |               |               |               |  |
| Cost of capital                                |               |                   | 18.00 %                                        |               |               |               |  |
| Investment in NWC                              |               |                   | 5.00 % of the predicted change in firm revenue |               |               |               |  |
| Initial investment in PPE                      |               |                   | \$7,000,000                                    |               |               |               |  |
| Annual Depreciation (5-year life w/no salvage) |               |                   | \$1,400,000                                    |               |               |               |  |
| <b>Solution</b>                                |               | <b>Cash Flows</b> |                                                |               |               |               |  |
| <b>Project Analysis</b>                        | <b>Year 0</b> | <b>Year 1</b>     | <b>Year 2</b>                                  | <b>Year 3</b> | <b>Year 4</b> | <b>Year 5</b> |  |
| Investment                                     | (7,000,000)   |                   |                                                |               |               |               |  |
| Revenue                                        |               | 146,850,000       | 154,192,500                                    | 161,902,125   | 169,997,231   | 178,497,093   |  |
| Variable cost                                  |               | (127,500,000)     | (133,875,000)                                  | (140,568,750) | (147,597,188) | (154,977,047) |  |
| Fixed cost                                     |               | (9,000,000)       | (9,000,000)                                    | (9,000,000)   | (9,000,000)   | (9,000,000)   |  |
| Depreciation                                   |               | (1,400,000)       | (1,400,000)                                    | (1,400,000)   | (1,400,000)   | (1,400,000)   |  |

|                                        |              |             |             |             |             |
|----------------------------------------|--------------|-------------|-------------|-------------|-------------|
| EBT (net operating income)             | 8,950,000    | 9,917,500   | 10,933,375  | 12,000,044  | 13,120,046  |
| Tax                                    | (4,475,000)  | (4,958,750) | (5,466,688) | (6,000,022) | (6,560,023) |
| Net operating profit after tax (NOPAT) | 4,475,000    | 4,958,750   | 5,466,688   | 6,000,022   | 6,560,023   |
| Plus: Depreciation expense             | 1,400,000    | 1,400,000   | 1,400,000   | 1,400,000   | 1,400,000   |
| Less: CAPEX                            | (7,000,000)  | —           | —           | —           | —           |
| Less: Change in NWC                    | (7,342,500)  | (385,481)   | (404,755)   | (424,993)   | 8,924,855   |
| Free cash flow                         | (14,342,500) | 5,507,875   | 6,461,932   | 6,975,029   | 16,884,878  |
| Net present value                      | \$ 9,526,209 |             |             |             |             |
| Internal rate of return                | 39.82%       |             |             |             |             |

## Project Risk Analysis

to be for next year? Construct a spreadsheet model and incorporate consideration for the uncertainty in future revenues to estimate the expected gross profit for the firm.

- b.** In the spring of 2010, Aggiebear Dog Snacks Inc. was estimating its gross profits (revenue less cost of goods sold) for 2011. The firm's CFO had recently attended a two-day seminar on the use of simulation and asked his analyst to construct a simulation model to make the estimate. To help guide the analyst, the CFO prepared the following table:

| Income Statement   | Variable Description                                                       |
|--------------------|----------------------------------------------------------------------------|
| Revenue            | Minimum = \$18 million; most likely = \$25 million; maximum = \$35 million |
| Cost of goods sold | 70% to 80% of revenue                                                      |

- i.** Construct a spreadsheet model for Aggiebear's gross profit.
- ii.** Use the information provided above to convert your spreadsheet model into a simulation model.
- iii.** Run your simulation model 10,000 iterations. What is the expected level of gross profit for 2011? What is the probability that the gross profit will fall below \$3.5 million?

**8 PROJECT RISK ANALYSIS USING SIMULATION** Rayner Aeronautics is considering a \$12.5 million investment that has an estimated project free cash flow (Project FCF) of \$2 million in its first year of operations. The project has a five-year life, and Rayner requires a return of 18% in order to justify making the investment.

- a.** What rate of growth in Project FCF for Years 2 through 5 is required for the project to break even (i.e., have an NPV = 0)?
- b.** Construct a simulation model for the investment opportunity to estimate the expected values of the Project FCF for Years 1 through 5. The first year's cash flow is normally distributed with an expected value of \$2 million and a standard deviation of \$1 million. Moreover, the rate of growth in Project FCF for Years 2 through 5 follows a triangular distribution with the following parameter estimates:

| Year | Most Likely Growth Rate (ML)             | Minimum Growth Rate                                        | Maximum Growth Rate                                  |
|------|------------------------------------------|------------------------------------------------------------|------------------------------------------------------|
| 2    | 40%                                      | $\frac{1}{2}$ of ML                                        | 2 times ML                                           |
| 3    | Actual growth rate for the previous year | $\frac{1}{2}$ the actual growth rate for the previous year | 2 times the actual growth rate for the previous year |
| 4    | Actual growth rate for the previous year | $\frac{1}{2}$ the actual growth rate for the previous year | 2 times the actual growth rate for the previous year |
| 5    | Actual growth rate for the previous year | $\frac{1}{2}$ the actual growth rate for the previous year | 2 times the actual growth rate for the previous year |

- c.** What are the expected NPV and IRR for the project, based on your simulation analysis from part b?

## PROBLEM 9

**MINI-CASE** CONOCOPHILLIPS GAS ACQUISITION PROJECT<sup>19</sup>

ConocoPhillips's (COP) Natural Gas and Gas Products Department (NG&GP) manages all of the company's activities relating to the gathering, purchase, processing, and sale of natural gas and gas liquids. Chris Simpkins, a recent graduate, was recently hired as a financial analyst to support the NG&GP department. One of Chris's first assignments was to review the projections for a proposed gas purchase project that were made by one of the firm's field engineers. The cash flow projections for the 10-year project are found in Exhibit P9.1 and are based on the following assumptions and projections:

- The investment required for the project consists of two components: First, there is the cost to lay the natural gas pipeline of \$1,200,000. The project is expected to have a 10-year life and is depreciated over seven years using seven-year MACRS.<sup>20</sup> Second, the project will require a \$145,000 increase in net working capital that is assumed to be recovered at the termination of the project.
- The well is expected to produce 900 MCF (thousand cubic feet) per day of natural gas during Year 1 and then decline over the remaining nine-year period (365 operating days per year). The natural gas production is expected to decline at a rate of 20% per year after Year 1.
- In addition to the initial expenditures for the pipeline and additional working capital, two more sets of expenses will be incurred. First, a fee consisting of 50% of the wellhead natural gas market price must be paid to the producer. In other words, if the wellhead market price is \$6.00 per MCF, 50% (or \$3.00 per MCF) is paid to the producer. Second, gas processing and compression costs of \$0.65 per MCF will be incurred.
- There is no salvage value for the equipment at the end of the natural gas lease.
- The natural gas price at the wellhead is currently \$6.00 per MCF.
- The cost of capital for this project is 15%.

## QUESTIONS

1. What are the NPV and IRR for the proposed project, based on the forecasts made above? Should Chris recommend that the project be undertaken? Why or why not? What reservations, if any, should Chris have about recommending the project to his boss?
2. Perform a sensitivity analysis of the proposed project to determine the impact on NPV and IRR for each of the following scenarios:
  - a. Best case: a natural gas price of \$8.00 and a Year 1 production rate of 1,200 MCF per day that declines by 20% per year after that.
  - b. Most likely case: a natural gas price of \$6.00 and a Year 1 production rate of 900 MCF per day that declines by 20% per year after that.
  - c. Worst case: a natural gas price of \$3.00 and a Year 1 production rate of 700 MCF per day that declines by 20% per year after that.

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<sup>20</sup> Modified Accelerated Cost Recovery System (MACRS) uses a shorter depreciable life for assets, thus giving businesses larger tax deductions and cash flows in the earlier years of the project life relative to those of straight-line depreciation.



**Exhibit P9.1****Analysis of the ConocoPhillips Gas Purchase Project**

| Year                                     | 0                     | 1                 | 2                 | 3                 |
|------------------------------------------|-----------------------|-------------------|-------------------|-------------------|
| Investment                               | \$ 1,200,000          |                   |                   |                   |
| Increase in NWC                          | 145,000               |                   |                   |                   |
| MACRS depreciation rate (7 year)         |                       | 0.1429            | 0.2449            | 0.1749            |
| Natural gas wellhead price (per MCF)     |                       | \$ 6.00           | \$ 6.00           | \$ 6.00           |
| Volume (MCF/day)                         |                       | 900               | 720               | 576               |
| Days per year                            |                       | 365               |                   |                   |
| Fee to producer of natural gas (per MCF) |                       | \$ 3.00           | \$ 3.00           | \$ 3.00           |
| Compression & processing costs (per MCF) |                       | 0.65              | 0.65              | 0.65              |
| <b>Cash Flow Calculations</b>            |                       |                   |                   |                   |
| Natural gas wellhead price revenue       |                       | \$1,971,000       | \$1,576,800       | \$1,261,400       |
| Lease fee expense                        |                       | 985,500           | 788,400           | 630,720           |
| Compression & processing costs           |                       | 213,525           | 170,820           | 136,656           |
| Depreciation expenses                    |                       | 171,480           | 293,880           | 209,880           |
| Net operating profit                     |                       | 600,495           | 323,700           | 284,184           |
| Less taxes (40%)                         |                       | (240,198)         | (129,480)         | (113,674)         |
| Net operating profit after tax (NOPAT)   |                       | 360,297           | 194,220           | 170,510           |
| Plus depreciation                        |                       | 171,480           | 293,880           | 209,880           |
| Return of net working capital            |                       |                   |                   |                   |
| <b>Project free cash flow</b>            | <b>\$ (1,345,000)</b> | <b>\$ 531,777</b> | <b>\$ 488,100</b> | <b>\$ 380,390</b> |

3. Do breakeven sensitivity analysis to find each of the following:

- Breakeven natural gas price for an NPV = 0.
- Breakeven natural gas volume in Year 1 for an NPV = 0.
- Breakeven investment for an NPV = 0.

4. Given the results of your risk analysis in Questions 2 and 3 above, would you recommend this project? Why or why not?

**PROBLEM 10** **MINI-CASE** **SOUTHWEST AIRLINES WINGLET PROJECT<sup>21</sup>**

As a low-fare airline, Southwest Airlines constantly focuses on ways to improve the efficiency of its operations and maintain a cost structure below that of its competition. In the spring of 2002, Scott Topping, the director of corporate, was approached by Aviation

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## Project Risk Analysis

| 4           | 5         | 6         | 7         | 8         | 9         | 10        |
|-------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 0.1249      | 0.0893    | 0.0893    | 0.0893    | 0.0445    |           |           |
| \$ 6.00     | \$ 6.00   | \$ 6.00   | \$ 6.00   | \$ 6.00   | \$ 6.00   | \$ 6.00   |
| 461         | 369       | 295       | 236       | 189       | 151       | 121       |
| \$ 3.00     | \$ 3.00   | \$ 3.00   | \$ 3.00   | \$ 3.00   | \$ 3.00   | \$ 3.00   |
| 0.65        | 0.65      | 0.65      | 0.65      | 0.65      | 0.65      | 0.65      |
| \$1,009,152 | \$807,322 | \$645,857 | \$516,686 | \$413,349 | \$330,679 | \$264,543 |
| 504,576     | 403,661   | 322,929   | 258,343   | 206,674   | 165,339   | 132,272   |
| 109,325     | 87,460    | 69,968    | 55,974    | 44,779    | 35,824    | 28,659    |
| 149,880     | 107,160   | 107,160   | 107,160   | 53,400    | —         | —         |
| 245,371     | 209,041   | 145,801   | 95,209    | 108,495   | 129,516   | 103,613   |
| (98,148)    | (83,616)  | (58,320)  | (38,083)  | (43,398)  | (51,806)  | (41,445)  |
| 147,223     | 125,425   | 87,480    | 57,125    | 65,097    | 77,710    | 62,168    |
| 149,880     | 107,160   | 107,160   | 107,160   | 53,400    | —         | —         |
|             |           |           |           |           |           | 145,000   |
| \$ 297,103  | \$232,585 | \$194,640 | \$164,285 | \$118,497 | \$ 77,710 | \$207,168 |

Partners Boeing (APB) regarding an innovative way to save on fuel costs—the installation of a new technology known as the Blended Winglet. The winglets, made of carbon-graphite, were designed for the Boeing 737-700 aircraft. Southwest currently had 142 planes of this model in its fleet.

The Blended Winglet system was developed by APB, a joint venture between Aviation Partners Inc. and The Boeing Company. The main purpose of the winglet was to reduce turbulence, leading to higher flying efficiency. As a result, the winglets provided three important benefits that allowed the airplane to extend its range, carry a greater payload, and save on fuel consumption. The winglets accomplished this by increasing the spread of the wings' trailing edge and creating more lift at the wingtips.

To complete his financial analysis, Scott had to verify potential costs and benefits as well as get the approval of the Maintenance Engineering Department, the Flight Operations Department, and the Facilities Department. However, Scott was well aware that regardless of the potential financial benefits, safety was the first priority. This process took several months due to the complexity of the project.

## Project Risk Analysis

After discussing the project with the requisite departments, Scott made the following estimates of the costs and benefits of the winglet system to Southwest:

- The winglets, which cost \$700,000 a pair, could be installed at an additional cost of \$56,000 per aircraft. Installation could be scheduled at each maintenance facility to coincide with regular maintenance. As a result, each aircraft was expected to experience downtime for only one extra day, at a cost of \$5,000.
- After considering the short- and long-term effects of the winglets, the Maintenance Engineering Department estimated that, on average, repair costs would average \$2,100 yearly per aircraft, due primarily to incidental damage.
- The increased wingspan was expected to allow each of Southwest's aircraft to fly up to 115 nautical miles further and to decrease fuel usage by 4% to 6%. This meant that Southwest could expect to save 178,500 gallons of jet fuel per airplane per year.
- Flight Operations<sup>22</sup> estimated that the additional lift capability provided by the winglets would reduce Southwest's costs of using restricted runways, with an estimated savings of \$500 per aircraft per year.
- The Facilities Department assessed the effect of the added wingspan on each of the 59 airports Southwest utilized in its current route structure. The department estimated that the necessary facilities modifications could be achieved at a onetime cost of about \$1,200 per aircraft.
- The Blended Winglet project qualified for accelerated tax write-off benefits under the Job Creation and Worker Assistance Act of 2002. With a marginal tax rate of 39% and using a seven-year depreciation schedule (see the table below), Southwest would be allowed to depreciate an additional 50% of the project in the first year.

| Depreciation Details |                    |                        |                         |                        |
|----------------------|--------------------|------------------------|-------------------------|------------------------|
| Year                 | Normal MARCS Table | Normal Table Times 50% | Year 1 (additional 50%) | Total (modified table) |
| 1                    | 14.29%             | 7.15%                  | 50.00%                  | <b>57.15%</b>          |
| 2                    | 24.49%             | 12.25%                 |                         | <b>12.25%</b>          |
| 3                    | 17.49%             | 8.75%                  |                         | <b>8.75%</b>           |
| 4                    | 12.49%             | 6.25%                  |                         | <b>6.25%</b>           |
| 5                    | 8.93%              | 4.47%                  |                         | <b>4.47%</b>           |
| 6                    | 8.92%              | 4.46%                  |                         | <b>4.46%</b>           |
| 7                    | 8.93%              | 4.47%                  |                         | <b>4.47%</b>           |
| 8                    | 4.46%              | 2.23%                  |                         | <b>2.23%</b>           |

<sup>22</sup> Southwest Airlines estimated the salvage value to be approximately 15% of the winglets' cost; however, this salvage value is very uncertain because this is a new technology with no historical data upon which to base an estimate. Scott felt that the winglets would definitely have some residual value, so 15% seemed like a reasonable salvage value to use based on the limited data available. In the analysis, the winglets should be depreciated to zero and the salvage value in Year 20 should be treated as taxable income.

## Project Risk Analysis

- The Blended Winglet project is expected to have a life of at least 20 years, at the end of which the winglets are anticipated to have a salvage value of \$105,000. Assume a jet fuel cost of \$0.80 per gallon and a cost of capital of 9.28% in your analysis. Items other than fuel are expected to escalate at a 3% rate. Conduct the analysis on a per-plane basis.

Evaluate the project by analyzing the following:

- a. Estimate the project's annual project free cash flow (FCF) for each of the next 20 years, as well as the initial cash outflow.
- b. Calculate the net present value (NPV) and internal rate of return (IRR) of the Blended Winglet project.
- c. What is the breakeven jet fuel cost for the project? What is the breakeven fuel savings in gallons for the project, assuming jet fuel costs \$0.80 per gallon?
- d. How sensitive is the Blended Winglet project's NPV to changing assumptions regarding expected future fuel costs and fuel savings? Use scenario analysis to analyze a best-case scenario (jet fuel price of \$1.10 per gallon and fuel savings of 214,000 gallons per year) and a worst-case scenario (jet fuel price of \$0.50 per gallon and fuel savings of 142,000 gallons per year).
- e. What potential risks and benefits do you see that are not incorporated into the quantitative analysis?
- f. What is the impact on the project's NPV or IRR if the winglets have no salvage value?
- g. Would you suggest Southwest Airlines undertake this project? Why or why not?

# An Introduction to Simulation Analysis and Crystal Ball

## What Is Simulation?

Simulation refers to the process of constructing a model that imitates a real-life situation: for example, the cash flows resulting from the operation of a power plant.

## Why Build Simulation Models?

When cash flows are determined from multiple sources of uncertainty, determining expected cash flows is incredibly complex. In these cases, simulation provides a tool that can be used to characterize the solution. Essentially, computer power is used to develop a distribution of possible solutions that reflects the underlying sources of uncertainty in the problem.

## What Is Crystal Ball Software?

Crystal Ball is an “add-in” software package that allows you to construct simulation models within Microsoft Excel spreadsheet software. Crystal Ball is a product of Decisioneering Inc.<sup>23</sup> Although we use Crystal Ball throughout this text, an alternative simulation add-in called @Risk is provided by the Palisade Corporation.<sup>24</sup>

## How Do I Build a Simulation Model Using Crystal Ball?

Excel models are **deterministic**, which means that the inputs are fixed (one value to one cell). You can see only one solution at a time. If you want to view alternative results, you need to manually change the inputs in the model. Crystal Ball software provides the user with the ability to make changes in the inputs to the model dynamically, such that many solutions to the model can be calculated and stored for later

<sup>23</sup> <http://www.crystalball.com/index.html>

<sup>24</sup> <http://www.palisade.com/>

analysis. To convert a deterministic spreadsheet model into a dynamic simulation model requires three steps:

- First, we must identify the inputs to the model that are subject to uncertainty and assign each a probability distribution. In Crystal Ball, these uncertain, or stochastic, inputs are called **assumptions**.
- Second, we must identify the particular output measures that are important to the analysis of the problem being analyzed. Crystal Ball refers to these as **forecasts**.
- Finally, once the assumptions and forecasts have been identified (we will talk about how in just a moment), we can start the simulation. This process entails selecting a value for each of the assumptions in the model, calculating the value of each of the forecast variables, and storing the result. This process is repeated for the number of **iterations** we specify.

The output of the simulation experiment consists of all the stored values of the forecast variables (e.g., NPV or IRR). The stored forecast variable data can be summarized using descriptive statistics such as the mean, median, and standard deviation—or by constructing a histogram or frequency distribution.

To illustrate the use of Crystal Ball in a very simple setting, consider the situation faced by WiseData Corp. The firm is evaluating an investment in a new product line that it expects to produce sales revenues for only one year. The firm's management is trying to forecast the gross profit that might be produced by the investment (as part of its analysis of project cash flows). Management's initial estimate of gross profit is \$20,000 and is found below in a static Excel model:

|                          | 2010                 |
|--------------------------|----------------------|
| Sales                    | \$100,000.00         |
| Less: Cost of goods sold | <u>\$(80,000.00)</u> |
| Gross profit             | <u>\$ 20,000.00</u>  |

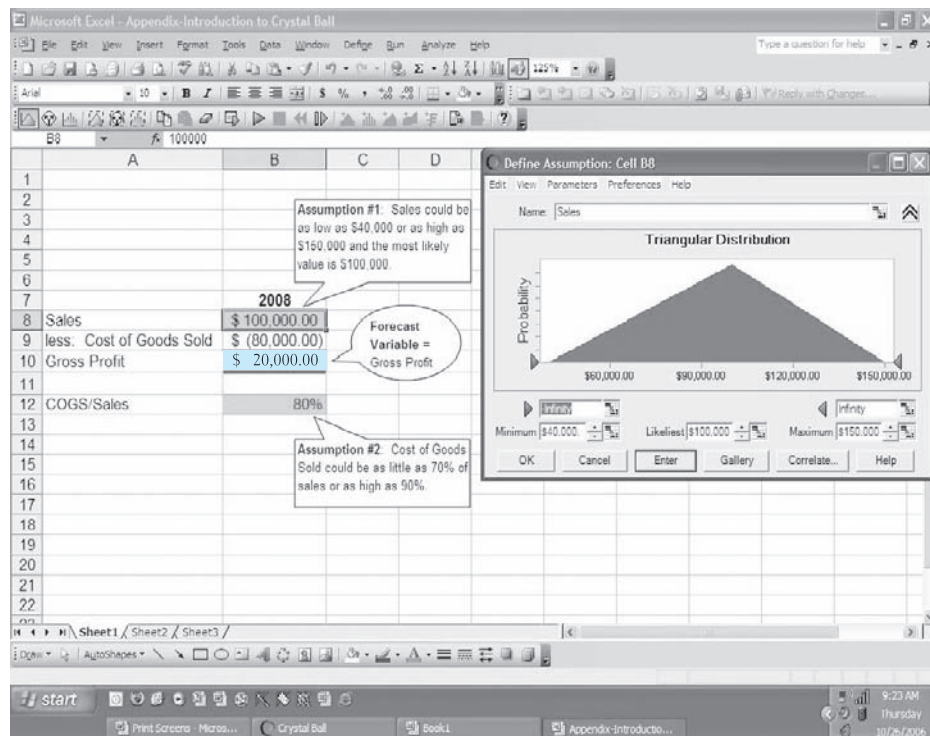
However, management realizes that both the level of sales and the cost of goods sold are not known with certainty, such that gross profit is also uncertain. To model gross profit using simulation to evaluate its underlying uncertainty, management has assessed as follows the uncertainty of sales and cost of goods sold:

| Variable (Assumption) | Description of Uncertainty                                                                   | Probability Distribution                                                                                   |
|-----------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Sales                 | Could be as low as \$40,000 or as high as \$150,000, but the most likely value is \$100,000. | Triangular with minimum value of \$40,000; most likely value of \$100,000; and maximum value of \$150,000. |
| Cost of goods sold    | Could be as little as 70% of sales or as much as 90%.                                        | Uniform distribution with a minimum value of 70% and a maximum value of 90%.                               |

## Project Risk Analysis

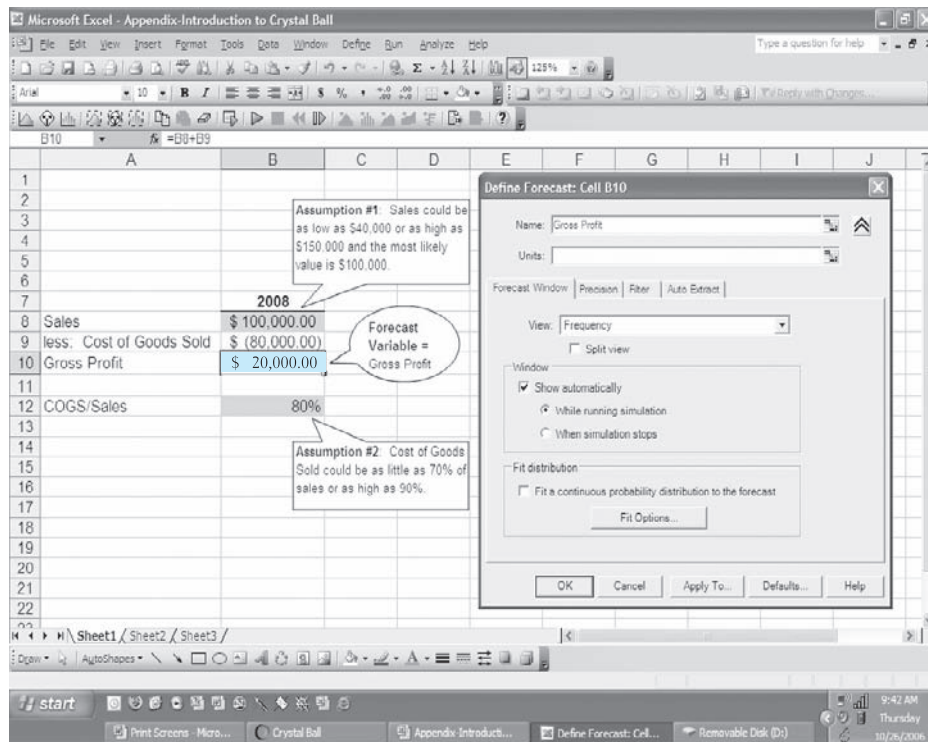
Note that the firm's management has identified the variables in the model that are subject to uncertainty, described the nature of the uncertainty (perhaps with the direct input of individuals within the company who are most familiar with the problem), and translated the description of underlying uncertainty into an appropriate probability distribution. Each of these steps requires skill and managerial judgment that can be developed over time with practice.

Crystal Ball adds icons to the spreadsheet toolbar that facilitate the process of defining assumptions and identifying forecasts. The screen containing the final model after placing the cursor on cell C8 (sales) and selecting the  and entering the minimum, most likely (likeliest), and maximum parameter estimates for the triangular distribution appears as follows:

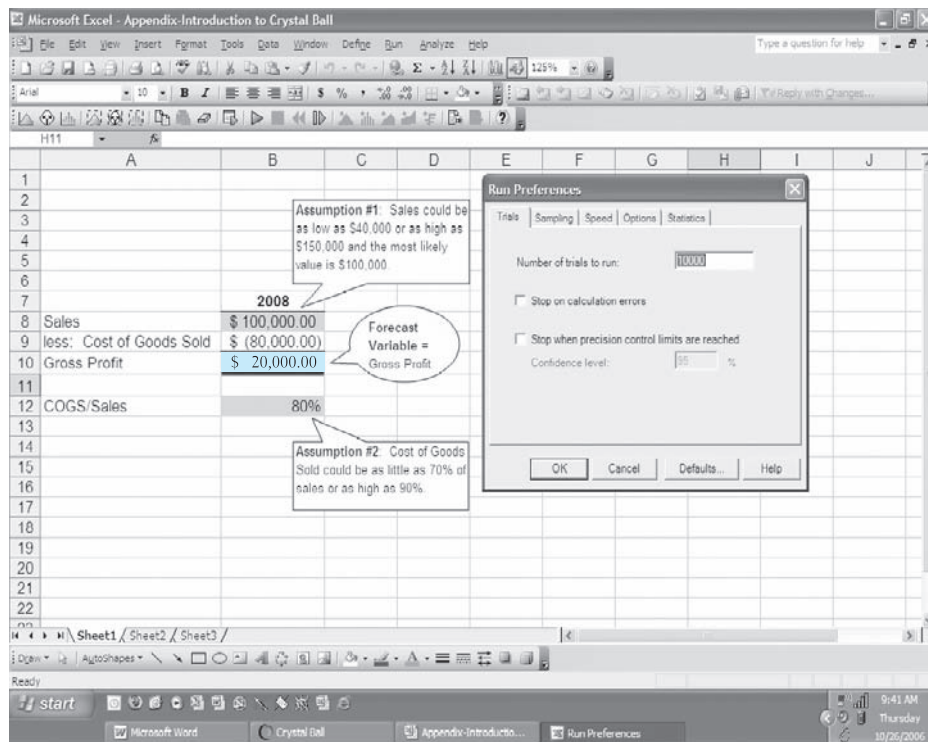


We assign a uniform distribution to cell B8 to make the cost of goods sold to sales ratio stochastic, following the same procedure we used in defining sales. Note that in the final model (above), the shaded cells identify the assumptions (i.e., random or stochastic variables), which include sales and cost of goods sold as a percent of sales, and the blue cell identifies the one output or forecast variable in the model that is also the object of the analysis. To define the forecast variable, we select the  in the toolbar. The Excel screen appears as follows:





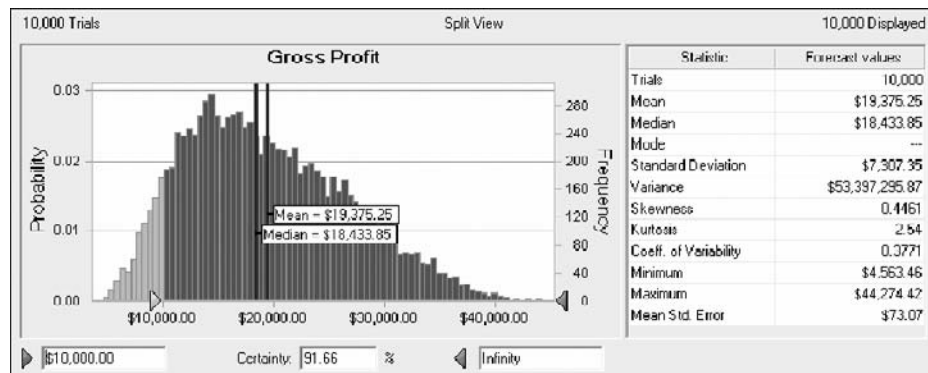
The final step before running the simulation involves selecting the  icon on the toolbar to define the number of iterations we wish to include in the simulation experiment. The Excel screen appears as follows:





## Project Risk Analysis

To run the simulation, we select the  icon on the toolbar. We ran the simulation 10,000 times (iterations), and Crystal Ball compiled the resulting 10,000 estimates of gross profit into the following histogram (frequency distribution):



Let's review the basic elements of the histogram of gross profit. First, this is the "Split View" of the results, which contains both the frequency distribution and statistics. (Crystal Ball offers other options to view the forecast output that we will not review here.) Next, there are 10,000 simulation trials and 10,000 iterations displayed in the chart. At the bottom of the frequency distribution are three values: \$10,000.00, 91.66%, and Infinity. The first number refers to the truncation point in the frequency distribution that appears above, and the "Certainty" of 91.66% is the percentage of the simulated gross profit values that were above \$10,000. In other words, there is a 91.66% chance that gross profit will be at least \$10,000 or an 8.34% chance that it will be less than \$10,000.